

Review

Legumes impacts and interactions with soil erosion

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ABSTRACT

Soil erosion is a major challenge for agroecosystem sustainability, especially when intense rainfall events correspond to limited soil cover. This review examines how herbaceous legumes influence erosion-related processes, focusing on affecting the cover-management factor (C) and K-factor related soil properties within RUSLE framework. Legume effect on soil erosion was affected by soil conditions, climate, species traits, management, phenological stage, and the reference system. Overall, legume-based systems performed better than bare soil and sole-crop systems in mitigating runoff and soil loss. Runoff reduction ranged from 8 to 75%; while soil or sediment loss decreased from 13 to 92% depending on the reference treatment and management context, including bare soil or sole crops, no-tillage with full-surface cover, legume strips, legume-grass mixtures, and crop phenological stages. C factor for legume (dried pulses and protein crops) is 0.32 in standard conditions but ranged from 0.21 to 0.23 for pure legumes covers, down to 0.015 in legume-grass mixes. Legumes also affected K-related properties by improving aggregation, root cohesion, pore structure, and infiltration, although direct K-factor changes were rarely quantified. Infiltration often improved relative to bare soil or fallow, although comparisons between legumes and grasses remained strongly context-dependent. Grasses and cereal cover usually offered faster canopy closure, greater biomass, and more persistent residues, whereas legumes contributed mainly through nitrogen inputs, residue quality, rhizosphere activity, root-channel formation and longer-term soil structure changes. Legumes can support erosion-control strategies; however, their most reliable contribution occurs in diversified systems, legume-grass mixes, and conservation practices maintaining continuous soil cover during erosive periods.

1. Introduction

Soil erosion is among the most urgent threats to global agroecosystem stability and food production. In Europe, the removal of an estimated 135 million m³ of sediments accumulated because of water erosion is likely to cost more than €2.3 billion per year in the EU and UK (Panagos et al., 2024). At the global scale, soil erosion contributes to soil depth reduction, nutrient and organic matter losses, and changes in soil biological functioning and biogeochemical cycling, with long-term consequences for agricultural productivity and agroecosystem resilience (Pimentel and Burgess, 2013; Quinton and Fiener, 2024).

Ellison (1947) and Meyer and Wischmeier (1969) defined soil erosion as “a process of detachment and transportation of soil materials by erosive agents”. Erosion involves two linked processes: soil particle detachment and transport. Soil particle detachment, the plucking of soil material from the soil surface where it is held by cohesion and inter-particle friction, has to be followed by the transport of the detached

material away from the site of detachment for erosion to occur. Both raindrop impact and surface water flows can detach and transport soil particles. In sheet and inter-rill erosion, raindrop impact is the main cause of detachment although raindrop splash and rain-impacted flows may be involved in transporting sediment over the land surface (Kinnell, 2022). In their 1969 study, Meyer and Wischmeier building on Ellison's work, proposed a simulation approach accounting for detachment potential and transport capacity of rainfall and runoff, as well as soil susceptibility to detachment and transport. Meyer and Wischmeier (1969) also emphasized that erosion simulation should consider seepage and subsurface flow, vegetation and crop residues, tillage, freezing and thawing, human-induced and natural disturbances, and topographic and microtopographic controls on runoff and rainfall exposure. In this context, conservation agriculture practices such as cover cropping, residue retention, and intercropping can reduce erosion by limiting raindrop impact, maintaining surface protection, slowing runoff, and improving soil structure (Kumawat et al., 2023).

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From Meyer and Wischmeier (1969)'s work, both physically based as well as conceptual and empirical models have evolved, which are still effectively used today for estimating and interpreting the erosion process.

Among the available models, the Universal Soil Loss Equation (USLE), developed by Wischmeier and Smith (1978), has become the most widely adopted framework for estimating soil erosion and serves as the foundation for several subsequent erosion prediction models. Within this framework, rainfall erosivity, defined as the capacity of rainfall and associated runoff to detach and transport soil particles-is quantified based on rainfall kinetic energy and intensity. Raindrop impact breaks down soil aggregates and detaches particles, whereas high-intensity rainfall enhances runoff generation when precipitation rates exceed the soil's infiltration capacity. The resulting overland flow on sloping surfaces further contributes to soil detachment and sediment transport (Brady and Weil, 2016). These processes are represented by the rainfall erosivity factor (R), originally introduced in the USLE (Wischmeier and Smith, 1978) and subsequently incorporated into the Revised Universal Soil Loss Equation (Renard et al., 1997), with updated methodologies for its estimation. Slope length and steepness are key topographic controls on runoff velocity, flow depth, and shear stress, which in turn influence erosion processes. In the RUSLE model, these effects are accounted for by the slope length (L) and slope steepness (S) factors, respectively. Soil properties influence the susceptibility of soil to detachment and transport by rainfall and runoff. In the RUSLE model, these effects are accounted for by the soil erodibility factor (K) (Wischmeier and Smith, 1978), which is traditionally estimated from soil texture, permeability, and organic matter content (Brady and Weil, 2016; Wischmeier and Smith, 1978). Vegetation cover and management practices affect the susceptibility of soil to erosion by reducing raindrop impact, slowing runoff, and limiting sediment transport. Over longer time scales, cover crops and residue retention may also improve soil structure, although this depends on species, soil type, and management practices (Brady and Weil, 2016; Kervroëdan et al., 2021). In the RUSLE model, these effects are represented by the cover-management factor (C) (Wischmeier and Smith, 1978). Land management practices can reduce soil erosion by modifying the flow path, velocity, and volume of surface runoff. Structural interventions such as contour farming, strip cropping, terracing, and grassed waterways decrease effective slope length and gradient, thereby limiting soil detachment and sediment transport over multiple growing seasons (Wischmeier and Smith, 1978). In the RUSLE model, these effects are represented by the support practice factor (P), with lower values corresponding to greater reductions in erosion potential. Owing to this multiplicative structure, reductions in any factor proportionally decrease the estimated soil loss.

In this context, herbaceous legumes have gained increasing attention because they can influence both soil surface protection and soil properties associated with erodibility. In Mediterranean environments, erosion control depends strongly on the temporal alignment between crop cover and periods of high rainfall erosivity, which may occur in summer and autumn or during humid transitional seasons depending on local rainfall regimes (Borrelli et al., 2016; Todisco et al., 2025). The agronomic flexibility of several legume species allows their integration into different crop-cycle windows, potentially improving this synchronization (Murphy-Bokern et al., 2017).

Through aboveground biomass production, canopy development, and residue deposition, legumes can reduce soil exposure to raindrop impact, slow surface runoff, and contribute to erosion mitigation, although their protective effect is often less persistent than that provided by grasses or cereal cover crops (Mwanamwenge et al., 1997; Ramirez-Garcia et al., 2015; Scavo et al., 2022). Belowground, legumes interact with rhizobia and arbuscular mycorrhizal fungi, enhancing nutrient cycling and microbial activity. Root exudates and microbial products may also promote aggregate formation and improve soil structural stability, potentially increasing infiltration and resistance to erosion processes (Blanco-Canqui and Ruis, 2020; Kabalan et al., 2022;

Wu et al., 2019).

Agronomic strategies such as green manuring, intercropping, conservation tillage, and species mixtures can further influence the erosion-mitigating performance of leguminous crops (Iderawumi and Kamal, 2022; Kumawat et al., 2022; Scavo et al., 2022). Consequently, legumes can be integrated as cover crops, intercrops, green manures, or main crops, although their effectiveness depends on the amount and timing of soil cover, residue persistence, root development, and management context (Kocira et al., 2020; Kumawat et al., 2022; Mohammed et al., 2021; Rodrigo-Comino et al., 2020; Scavo et al., 2022).

Despite these recognized benefits, important knowledge gaps remain. Existing studies often focus on individual erosion-related indicators rather than providing an integrated assessment of how legumes influence the different processes represented by the RUSLE factors. Moreover, most available evidence is derived from comparisons with bare soil, making it difficult to distinguish legume-specific effects from the general benefits of vegetation cover (Gabriel et al., 2021; García-González et al., 2018; Naderi Marangalo et al., 2025; Ramirez-Garcia et al., 2015; Rodrigo-Comino et al., 2020; Sastre et al., 2018; Yao et al., 2025). Comparative analyses against well-performing grass and cereal covers remain limited, and the relative contribution of aboveground protection mechanisms and belowground modifications of soil properties is still poorly quantified across different climates, soil types, and management systems. Furthermore, the extent to which legumes affect specific RUSLE components, particularly the cover-management factor (C) and the soil erodibility factor (K), has not been systematically synthesized.

This review addresses these gaps by examining the above- and belowground mechanisms through which herbaceous legumes influence erosion-related processes and by interpreting their effects within the RUSLE framework focusing primarily on how leguminous species can influence C factor and soil properties associated with K, particularly through canopy development, residue cover, root traits, aggregation, and infiltration.

Specifically, the review: i) analyse how legume morphology and development dynamics influence erosion-related processes; ii) identify crop-management practices that can enhance the erosion-control functions of legumes in semi-arid, temperate and Mediterranean agroecosystems; iii) evaluate agronomic practices relevant to erosion control based on current literature evidence; iv) compare the effectiveness of legume-grass or legume-cereal mixes with pure leguminous stands; v) provide practical recommendations and research priorities for improving the use and the parameterization of legume in erosion-control strategies.

2. Methods

2.1. Conceptual basis

As shown in Fig. 1, water erosion under bare soil conditions is initiated by the direct impact of high-energy raindrops, which trigger splash erosion and particle detachment. When rainfall intensity exceeds infiltration capacity, surface runoff (water that flows over the land surface) develops and transports detached particles downslope. As runoff concentrates downslope, its erosive power progressively increases, resulting in enhanced soil detachment, sediment transport, and surface degradation. (Brady and Weil, 2016).

Fig. 1 illustrates how vegetation acts as a natural barrier to erosive processes, reducing both surface runoff and sediment delivery (Kervroëdan et al., 2021). The plant canopy intercepts rainfall and reduces the kinetic energy of impacting raindrops (impacting raindrop, kinetic energy reduction, lower detachment capacity), thereby diminishing their ability to detach soil particles and limiting the distance over which particles are ejected (lower transport capacity) (Kervroëdan et al., 2021; Wischmeier and Smith, 1978). At the soil surface, the presence of plant residues and stems increases soil surface hydraulic resistance

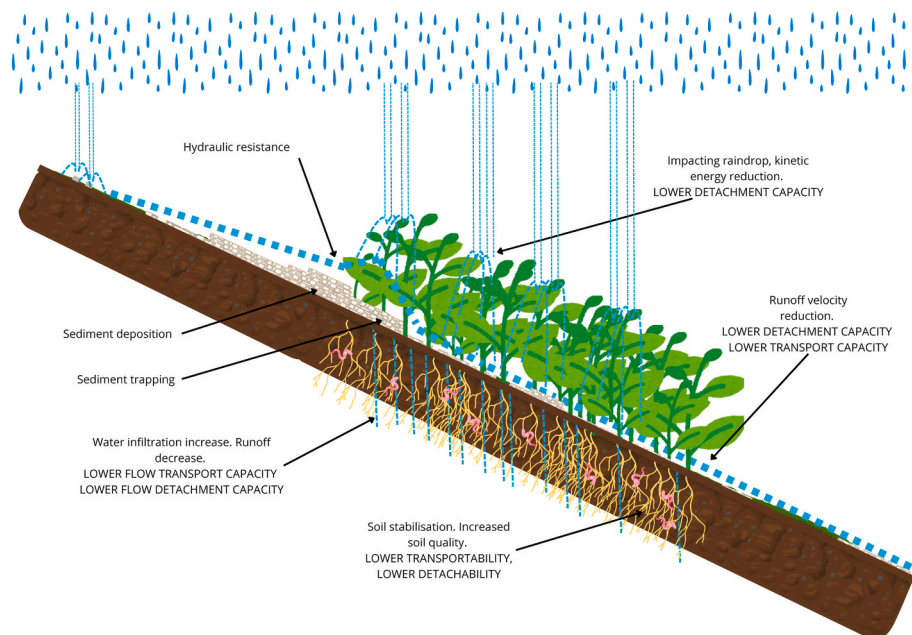


Fig. 1. Schematic illustration of the main hydrological and mechanical processes involved in soil water erosion and the mitigating role of vegetation cover on a sloping surface. Vegetation canopy reduces the kinetic energy of raindrops impacting the soil surface and soil particle detachment, while stems and surface roughness decrease runoff velocity and transport capacity. Root systems enhance soil stabilization and infiltration, promoting sediment trapping, deposition, reducing soil detachability and transportability by both raindrop impact and surface runoff.

(resistance to water flow over the land surface), slowing down runoff velocity and reducing its erosive power (lower detachment and transport capacity), promoting infiltration (Brady and Weil, 2016;

Kervroëdan et al., 2021). Stems also promote sediment deposition and trapping, further contributing to erosion control and soil conservation along the slope (Kervroëdan et al., 2021). In natural conditions,

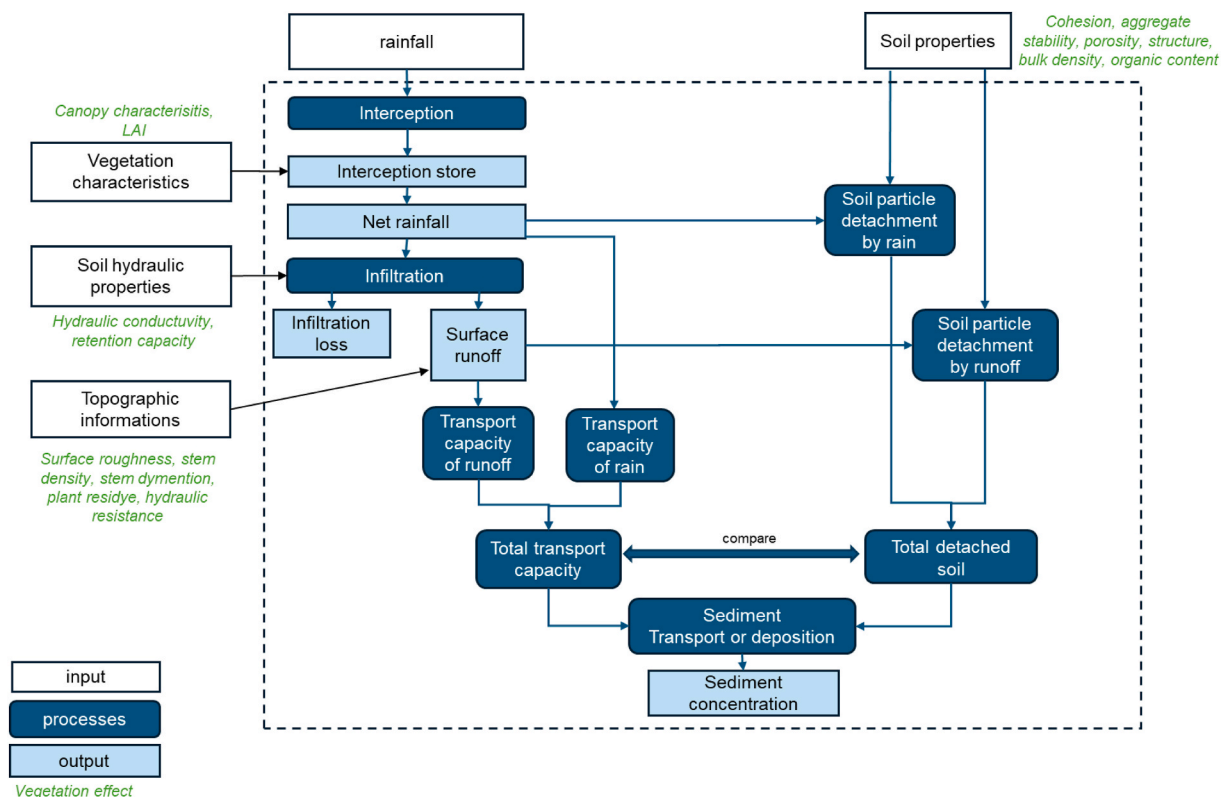


Fig. 2. Conceptual flowchart of soil erosion processes under vegetated conditions which also apply to legume crops and legume covered soils, adapted from Ellison (1947), Meyer and Wischmeier (1969), and Ramsankaran et al. (2013). The diagram illustrates the main interactions and feedbacks among rainfall characteristics, surface runoff generation, soil detachment, transport and deposition processes. Vegetation structural and functional traits influence erosion dynamics by modifying rainfall interception, runoff velocity, sediment transport capacity, and soil properties such as aggregation, porosity, and infiltration, ultimately affecting erosion rates at the hillslope scale.

fine-grained soils generally occur as aggregates (individual particles bound together into larger structural units) rather than as individual particles. Root systems enhance soil physical properties such as aggregate stability (the ability of soil aggregates to resist disintegration when subjected to external forces), soil structural stability (ability of soil aggregates and pore networks to maintain their arrangement), soil cohesion (internal force that binds soil particles together, providing resistance to aggregate separation). These characteristics increase the resistance of soil aggregates, reducing their susceptibility to detachment and transport by erosive agents; moreover, they contribute to improving infiltration rates (water infiltration increase), and reducing runoff volume (runoff decrease), thereby lowering the flow transport capacity and the potential for both detachment and transport (Brady and Weil, 2016; Wischmeier and Smith, 1978). While the mechanisms described above apply broadly to herbaceous vegetation, their expression may differ between plant functional groups. For example, annual legumes such as faba bean (*Vicia faba* L.) and common vetch (*Vicia sativa* L.) may provide short-term soil protection through canopy expansion (Mwanamwenge et al., 1997; Ramirez-Garcia et al., 2015), whereas grasses and cereal covers often provide denser fibrous root mats and more persistent residues (Blanco-Canqui and Ruis, 2020; Ramirez-Garcia et al., 2015; Van Eerd et al., 2023). Perennial or tap-rooted legumes such as alfalfa (*Medicago sativa* L.) and common sainfoin (*Onobrychis viciifolia* Scop.) may instead influence infiltration and soil stability through deeper root systems and longer-term changes in soil structure, while root-derived macro-pores have been shown especially for alfalfa and decayed root systems (Guo et al., 2019; Qu et al., 2025; Sastre et al., 2018; Yüsek and Yüsek, 2015).

Fig. 2 schematically represents the rainfall-runoff-soil erosion system in the presence of vegetation, highlighting the key input drivers and the associated process outputs. (Ellison, 1947; Meyer and Wischmeier, 1969; Ramsankaran et al., 2013). This review discusses the main variables, the key processes, and the physical interconnections between them, with particular attention to the influence of legumes on these relationships. The analysis emphasizes how leguminous plants affect soil stability, erosion dynamics, and the overall interactions between biotic and abiotic components within the system.

RUSLE is a widely used empirical model for estimating average annual soil loss from rainfall and surface runoff. In the RUSLE framework, soil loss (A) is expressed as the product of five factors: rainfall-runoff erosivity (R), soil erodibility (K), slope length and steepness (LS), cover-management (C), and support practices (P) (Renard et al., 1997; Wischmeier and Smith, 1978). Owing to its relative simplicity, empirical modelling remains attractive from a practical and operational perspective (Gao et al., 2012; Gessesse et al., 2015). As a multiplicative model, RUSLE is highly sensitive to reductions in individual factors; e.g. when one factor assumes very low values, the resulting estimated soil loss is correspondingly minimized.

Within this framework, vegetation effects are represented most directly through the cover-management factor (C), while longer-term changes in soil structure, aggregation, infiltration, and organic matter may influence soil properties associated with erodibility (K-related properties). The C factor explicitly represents the protective role of vegetation and management practices and allows temporal variability in soil cover to be incorporated, as it is commonly calculated as a weighted average of cover conditions throughout the agricultural year. It therefore captures the influence of erosion control practices cover crops, residue management, crop rotations, and reduced soil disturbance, that protect the soil from erosive forces and reduce erosion potential. The timing and seasonal dynamics of vegetation cover are closely linked to the temporal distribution of climatic forcing, which in turn interacts with the soil's inherent susceptibility to erosion as quantified by the K

factor (Ashiagbor et al., 2013; Wischmeier and Smith, 1978). In this review, the RUSLE framework was used as an analytical lens to interpret how herbaceous legumes, through their canopy development, root traits, and management, influence erosion-related processes, particularly those represented by the C and K factors.

2.2. Paper review and screening

This manuscript presents a review of how herbaceous legumes influence soil erosion-related processes. Given the marked heterogeneity of the available literature in terms of design, timescale, climate, soil conditions, and reference systems, the objective was not to derive pooled estimates, but to identify recurring mechanisms and context-dependent responses. Only studies addressing herbaceous legumes in relation to soil erosion or closely related processes, such as runoff, soil loss, infiltration, and aggregate stability, were considered.

The review was organized around the main erosion-related processes. The main focus of the review is around the cover-management factor (C), soil erodibility factor (K), and rainfall erosivity factor (R) referred to the Revised Universal Soil Loss Equation (RUSLE) framework. The literature search was conducted in Scopus and Web of Science between 25 February 2026 and 26 March 2026. Nine search queries were developed to cover the principal domain addressed in the review. In the first query the terms “legume” and “soil erosion” were not included, instead in all other search strings “legume” and “soil erosion” were included combined with additional keywords related to canopy development, runoff, infiltration, aggregate stability, mycorrhizae, root exudates, green manure, and RUSLE-related parameters. The full search strings and database-specific filters are reported in Table 1. The papers were selected after applying certain filters to the databases, such as language (English), the time period 1995–2026, and the relevant scientific fields, including: “Environmental Science”, “Agricultural and Biological Sciences”, “Earth and Planetary Sciences”, “Multidisciplinary” for Scopus; “Agriculture”, “Water Resources”, “Environmental Sciences Ecology”, “Plant Sciences”, “Science Technology Other Topics”, “Microbiology” for Web of Science. The scientific fields filter is used depending on the search query and depending on the database. The research procedure is represented in Fig. 3.

Records retrieved from Scopus and Web of Science were screened through a multi-step selection procedure. Because the same study could be retrieved by multiple queries or by both databases, articles retained at the title and abstract stage were counted as a unique entry only. When a study had already been selected in a previous query or database, it was not counted again in subsequent query-level totals. Titles and abstracts were screened and only studies satisfying the following criteria were selected: (i) focused on herbaceous leguminous species; (ii) addressed soil erosion processes or closely related mechanisms, including runoff generation, sediment loss, infiltration, soil aggregation, soil structural stability, or erosion-related crop management; and (iii) provided sufficient environmental, agronomic, or experimental context. Studies conducted in temperate, semi-arid, and Mediterranean environments were considered, although mechanistic studies from other conditions were also retained when they provided direct and transferable insight into legume-erosion relationships.

After title and abstract screening, 96 articles were retained for full-text assessment, including 57 from Scopus and 39 from Web of Science (Table 1). Full texts were then evaluated for relevance and consistency with the aims of the review. At this stage, 17 articles were excluded because they fell outside the climatic or agronomic scope of the manuscript, were not sufficiently relevant to the specific relationship between legumes and soil erosion, or could not be assigned to the process-based synthesis framework used in the review. This resulted in

Table 1

Search strategy and screening outcomes for the literature review.

The table reports the nine search queries used in Scopus and Web of Science, together with the corresponding database-specific search strings, post-filter records, and the number of unique articles retained after title/abstract screening and full-text assessment. Query-level counts refer to newly retained unique studies only; records already selected in previous queries or from the other database were not counted again.

Search Query Scopus	Search query Web of Science	Records retrieved in Scopus	Records retrieved in Web of Science
ALL ((“rainfall erosivity” OR “rainfall intensity” OR “R-factor”) AND europe AND legume AND rusle) AND (LIMIT-TO (SUBJAREA,“ENVI”) OR LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	(legume AND “soil erosion” AND europe) OR (“Rainfall erosivity” AND R-factor AND “rainfall intensity”) (All Fields) AND 1994 (Exclude - Publication Years) AND English (Languages) AND Agriculture or Water Resources or Environmental Sciences Ecology (Research Areas)	38 Post filter 36 Articles retained after title and abstract screening: 2 Articles retained after full-text screening: 2	60 Post filter 53 Articles retained after title and abstract screening: 2 Articles retained after full-text screening: 2
ALL (Legume AND (morphology OR “Canopy structure”) AND “soil erosion” AND runoff) AND PUBYEAR >1994 AND PUBYEAR <2027 AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	(legume AND “soil erosion” AND runoff) OR (morphology AND “Canopy structure”) (All Fields) AND 1994 or 1993 or 1991 or 1990 or 1986 or 1988 (Exclude - Publication Years) AND Agriculture or Plant Sciences or Environmental Sciences Ecology (Research Areas) AND English (Languages)	394 Post filter 360 Articles retained after title and abstract screening: 10 Articles retained after full-text screening: 10	274 Post filter 186 Articles retained after title and abstract screening: 5 Articles retained after full-text screening: 4
ALL ((legume AND “canopy development”) AND (“soil cover” OR “soil erosion”)) AND PUBYEAR >1994 AND PUBYEAR <2027 AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	(legume AND “soil erosion”) AND (“Ground cover” OR “soil cover”) AND 1985 or 1987 or 1986 or 1991 or 1992 (Exclude - Publication Years) AND Agriculture or Environmental Sciences Ecology or Plant Sciences (Research Areas) AND English (Languages)	66 Post filter 59 Articles retained after title and abstract screening: 3 Articles retained after full-text screening: 3	47 Post filter 34 Articles retained after title and abstract screening: 9 Articles retained after full-text screening: 7
ALL ((legume AND “green manure” AND “soil loss” AND “soil erosion” AND “soil fertility”) AND PUBYEAR >1994 AND PUBYEAR <2027 AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	(legume AND “soil erosion”) AND (“green manure” OR soil fertility) (All Fields) AND 1985 or 1993 or 1994 (Exclude - Publication Years) AND English (Languages) AND Agriculture or Environmental Sciences Ecology or Science Technology	147 Post filter 131 Articles retained after title and abstract screening: 8 Articles retained after full-text screening: 7	98 Post filter 79 Articles retained after title and abstract screening: 6 Articles retained after full-text screening: 2

Table 1 (continued)

Search Query Scopus	Search query Web of Science	Records retrieved in Scopus	Records retrieved in Web of Science
ALL ((rusle AND “C-factor” OR “canopy cover” OR “cover crops”) AND legume AND (“soil erosion” OR “soil health”)) AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	Other Topics (Research Areas) (rusle AND “C-factor” OR “canopy cover” OR “cover crops”) AND legume (“soil erosion” OR “soil health”) (All Fields) AND 1986 or 1992 or 1993 or 1994 (Exclude - Publication Years) AND English (Languages) AND Agriculture or Environmental Sciences Ecology or Science Technology	141 Post filter 134 Articles retained after title and abstract screening: 6 Articles retained after full-text screening: 5	236 Post filter 201 Articles retained after title and abstract screening: 4 Articles retained after full-text screening: 2
ALL ((legume AND “soil erosion”) AND (amf OR mycorrhizal) AND “aggregate stability” AND soil loss) AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”) OR LIMIT-TO (SUBJAREA, “MULT”) AND (LIMIT-TO (LANGUAGE, “English”))	(legume AND soil erosion) AND (mycorrhizal OR soil stability) AND Agriculture or Environmental Sciences Ecology or Plant Sciences (Research Areas) AND 1986 or 1989 or 1990 or 1991 or 1992 or 1993 or 1994 (Exclude - Publication Years) AND English (Languages)	272 Post filter 257 Articles retained after title and abstract screening: 7 Articles retained after full-text screening: 6	83 Post filter 63 Articles retained after title and abstract screening: 9 Articles retained after full-text screening: 8
ALL ((Legume AND “root exudate”) AND (“soil erosion” AND aggregate)) AND PUBYEAR >1996 AND PUBYEAR <2027 AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	Root exudates AND (legume AND “soil erosion” OR Aggregate) AND Agriculture or Plant Sciences or Microbiology (Research Areas) AND 1993 or 1992 or 1991 or 1990 or 1987 or 1986 (Exclude - Publication Years) AND English (Languages)	239 Post filter 218 Articles retained after title and abstract screening: 4 Articles retained after full-text screening: 3	133 Post filter 83 Articles retained after title and abstract screening: 1 Articles retained after full-text screening: 1
“soil erosion” AND (Legume AND root reinforcement AND shear) AND PUBYEAR >2005 AND PUBYEAR <2027 AND (LIMIT-TO (SUBJAREA, “ENVI”) OR LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO (SUBJAREA, “EART”)) AND (LIMIT-TO (LANGUAGE, “English”))	“soil erosion” AND (legume root) AND 1993 or 1992 or 1991 or 1990 or 1987 or 1986 (Exclude - Publication Years) AND English (Languages) AND Agriculture or Water Resources or Plant Sciences (Research Areas)	113 Post filter 98 Articles retained after title and abstract screening: 3 Articles retained after full-text screening: 3	57 Post filter 42 Articles retained after title and abstract screening: 2 Articles retained after full-text screening: 1
legume AND (“water infiltration” AND “soil erosion”) AND root AND (LIMIT-TO (SUBJAREA, “AGRI”) OR LIMIT-TO	legume AND (“water infiltration” OR “soil erosion” AND root) AND 1987 or 1991	460 Post filter 434 Articles retained after title	122 Post filter 100 Articles retained after title

(continued on next page)

Table 1 (continued)

Search Query Scopus	Search query Web of Science	Records retrieved in Scopus	Records retrieved in Web of Science
(SUBJAREA,"ENVI") OR LIMIT-TO (SUBJAREA,"EART") AND (LIMIT-TO (LANGUAGE, "English"))	or 1994 or 1993 or 1992 (Exclude - Publication Years) AND English (Languages) AND Environmental Sciences Ecology or Agriculture or Plant Sciences (Research Areas)	and abstract screening 14 Articles retained after full- text screening: 12	and abstract screening 1 Articles retained after full- text screening: 1
Overall totals		Articles retained after title and abstract screening 57 Articles retained after full- text screening: 51	Articles retained after title and abstract screening 39 Articles retained after full- text screening: 28

79 primary studies, of which 51 were retrieved from Scopus and 28 from Web of Science.

Additional documentation was submitted subsequent to consultation with experts in the field. These were not treated as part of the screened primary studies, but were used to strengthen specific sections of the manuscript. They included: (i) historical or mechanistic papers relevant to erosion dynamics, vegetated soil processes, soil physical processes, and rhizosphere interactions; and (ii) authoritative sources (e.g.

manuals) on legumes and agronomic management. This corpus comprises 52 papers. The final research corpus considered in the manuscript therefore comprised 131 references.

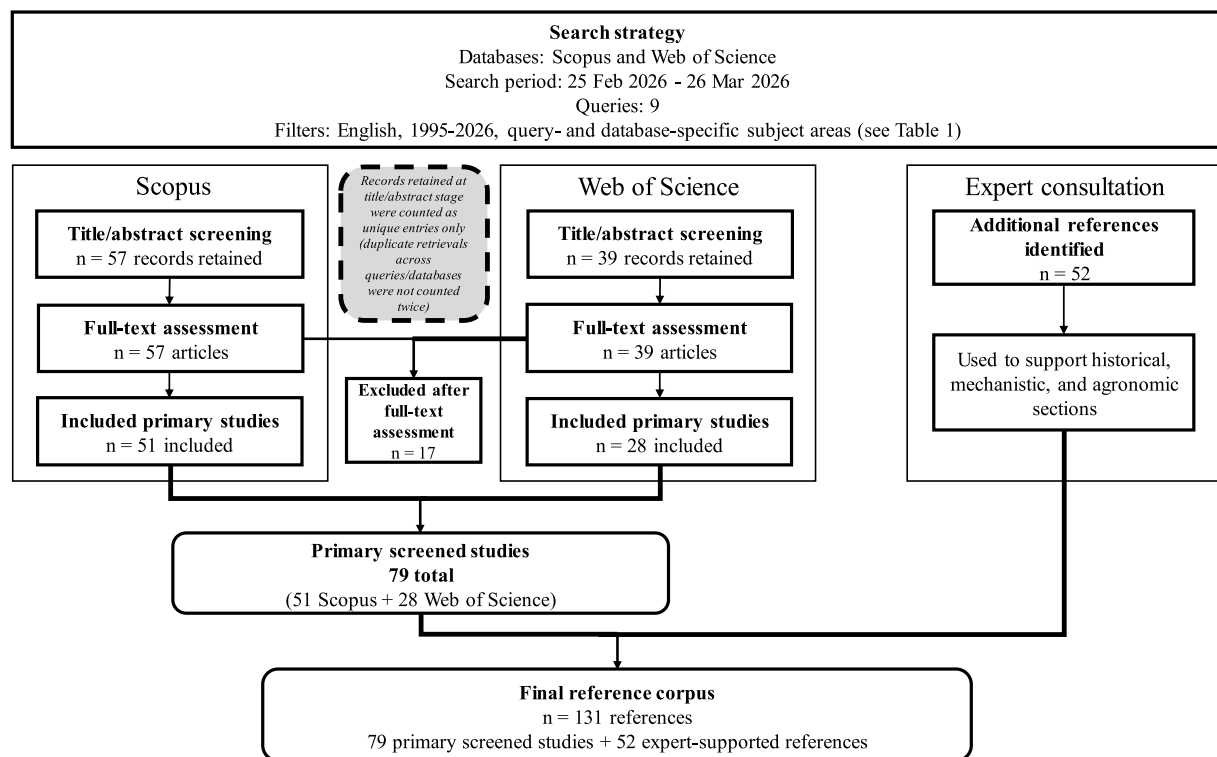
In line with the narrative and process-oriented nature of the review, the synthesis focused on mechanistic interpretation and context-dependent responses rather than generalized pooled estimates. Evidence was interpreted with explicit consideration of climate, soil conditions, cropping system, management intensity, and reference treatment.

3. Aerial part

3.1. Legumes canopy and ground cover (C factor)

According to Wischmeier and Smith (1978), the C factor varies seasonally with crop development stages, residue management and is crucial for assessing the protective role of vegetation and residues against raindrop impact and runoff. By definition, the C factor is 1 for bare soil and progressively decreases as vegetation cover increases (Ashiagbor et al., 2013; Wischmeier and Smith, 1978). According to Panagos et al. (2015b), legumes are classified under the "dried pulses and protein crops" category and are assigned a base C value (C_{crop}) of 0.32.

Experimental evidence supports the variability of C values among legume species. For instance, in a two-year field study conducted in a Sicilian vineyard, faba bean (*Vicia faba* L.) and a faba bean - common vetch (*Vicia sativa* L.) mixture exhibited C values of 0.21 and 0.23, respectively among the highest recorded for cover crops in that trial (Novara et al., 2011). By contrast, in the same conditions, the mixture of subterranean clover (*Trifolium subterraneum* L.) with grasses such as red fescue (*Festuca rubra* L.) and sheep fescue (*Festuca ovina* L.) reached the lowest C values, from 0.06 to 0.015. This suggests species composition and growth habit can strongly influence canopy effectiveness and soil protection (Novara et al., 2011). Overall, legumes can influence the C



Narrative, process-oriented synthesis focused on RUSLE-related dimensions (C, K, and R factors).

Fig. 3. Workflow of the literature screening and reference selection process, including database searches in Scopus and Web of Science, full-text assessment, expert-supported references, and the final corpus used for the narrative synthesis.

factor directly by increasing canopy and ground cover, and indirectly when integrated into cover crop mixtures, green manure systems, or intercropping practices that modify residue cover and soil surface protection. However, the available C-factor values indicate that their effectiveness is strongly dependent on species composition, cover persistence, growth habit, and management context.

3.2. Development, morphology, and impact on erosion of the legume canopy

Canopy effectiveness against soil erosion depends on planting season, biomass quality, and crop residues (Haruna et al., 2020). In an Australian study, under field conditions in a short season Mediterranean environment, Mwanamwenge et al. (1997) compared canopy development among winter-sown grain legumes, reporting that faba bean was the fastest in developed green area index (GAI) followed by field pea (*Lathyrus oleraceus* Lam.). Faba bean reached the highest GAI development rate of 0.091 d^{-1} , approximately 40% faster than field pea (0.061 d^{-1}) and narbon bean (*Vicia narbonensis* L.) (0.060 d^{-1}), and over four times faster than chickpea (*Cicer arietinum* L.) (0.022 d^{-1}), enabling earlier canopy closure and more efficient light interception during the critical early growth stages. The Green Area Index (GAI) refers to the one-sided area of photosynthetically active green tissues per unit ground surface, and differs from the Leaf Area Index (LAI), which also includes senescent foliage (Jiang et al., 2022). However, under Central European conditions, Neugschwandtner et al. (2019) found that winter wheat (*Triticum aestivum* L.) exhibited higher soil coverage than winter faba bean in one of the two years of their trial, and consistently more than winter field pea in both years. In Spain, under Mediterranean climate, (Ramirez-Garcia et al. (2015) quantitatively assessed canopy development among five cover crop species, reporting that common vetch reached up to 96% soil coverage under mild conditions on the first year, but only 47% in colder temperatures on the second year, demonstrating delayed canopy development and lower biomass accumulation compared to grasses. A rainfall-simulation field experiment on gently sloping farmland in the Loess Plateau further showed that soybean (*Glycine max* L.) growth increased surface protection during the crop cycle, with vegetation cover rising from 10% at seeding to 73% at ripening and plant height increasing from 21 to 68 cm. Over the same period, total runoff volume and soil loss decreased by 8–75% and 13–92%, respectively, relative to bare soil, with the strongest erosion reduction generally observed from final flowering to grain formation, particularly at the grain-forming stage (Yao et al., 2025). In a three-year semi-arid field trial conducted in Turkey, common sainfoin (*Onobrychis viciifolia* Scop.) grew slowly at the beginning of the 2006 growing period, canopy cover increased from 14% in March to 36% in July. Similar trends were observed in the following years, rising from 32% to 73% between March and July 2007, and from 50.5% to 77.7% over the same period in 2008, confirming its capacity to rapidly establish dense ground cover. Although natural vegetation composed of species such as Jerusalem thorn (*Paliurus spina-christi* Mill.), Bermuda grass (*Cynodon dactylon* (L.) Pers.), *Achillea* spp., and *Astragalus* spp. Initially provided higher surface cover (67–73%), common sainfoin approached its protective performance by the third year, recording the lowest sediment concentration in runoff in 2008 (1.76 g L^{-1} under sainfoin vs. 1.90 g L^{-1} under natural cover and 3.82 g L^{-1} on bare soil; all values refer to the year 2008) (Yüksek and Yüksek, 2015).

In a ten-year experiment conducted on irrigated fields in central Spain, vetch (*Vicia spp.*) provided moderate soil cover (44.4%), whereas barley (*Hordeum vulgare* L.) achieved 61.7% on average (Gabriel et al., 2021). The probability of extreme drought conditions (surface soil water content $<50 \text{ g kg}^{-1}$) was reduced from 55% on bare fallow land to 43% with barley, whilst vetch showed a similar trend to fallow land. In

Table 2

Integrated overview of the main mechanisms through which legumes may contribute to soil erosion control. The table summarizes the canopy, rhizosphere, soil-structure, hydrological, and management-related pathways discussed in the review. For each erosion-control driver, the expected effect is reported together with the main mechanism and the conditions that may strengthen or limit its contribution. Symbols indicate the overall interpretation of the available evidence: “+” = generally positive effect; “±” = context-dependent effect; “-” = no clear or negative effect.

Erosion control driver	Effects	Reason
Canopy cover (C factor)	+	Greater canopy and ground cover reduce raindrop impact and lower the C factor, with dense legume-grass mixtures generally providing stronger soil protection than pure legume stands.
Canopy development	±	Canopy development varies among legume species and management conditions; early-developing legumes can improve short-term soil protection, whereas delayed canopy closure may reduce protection compared with faster-establishing cereal or grass covers.
Synchrony with rainfall erosivity	±	Legume-based erosion control is most effective when canopy development and residue cover coincide with periods of high rainfall erosivity; poor synchrony leaves the soil exposed during high-risk periods.
Soil erodibility (K-related properties)	+	Root-driven improvements in soil structure, aggregation, infiltration, and biological activity can reduce K-related erodibility proxies, although responses depend on soil type and management.
Root mechanical traits	±	Legume roots can enhance soil cohesion and infiltration through physical reinforcement, taproot channels, and macropore formation; however, effects vary with root architecture, root decay, soil moisture, and pore occupation by dense live roots.
Water infiltration	±	Legume roots can improve infiltration and water retention through aggregation, macroporosity, and preferential flow, but responses are strongly mediated by species identity, root architecture, phenological stage, soil properties, and comparator type.
Symbiosis	±	Rhizobia and AMF can support nutrient cycling, microbial activity, and soil aggregation, thereby contributing indirectly to soil structural stability and resistance to erosion; direct erosion outcomes remain context-dependent.
Biochemical interaction	±	Root exudates, mucilage, organic acids, and microbially derived compounds can influence rhizosphere activity, nutrient mobilization, and aggregate stability, with indirect and context-dependent implications for soil detachability.
Conservation tillage	±	Conservation tillage can improve soil structure, residue retention, infiltration, and biological activity, especially when combined with legume cover crops; however, without sufficient surface cover it may promote sealing, compaction, or increased runoff.
Cover crops	±	Legume cover crops can improve fertility, microbial activity, aggregation, and erosion control, but rapid residue decomposition may limit persistent protection; legume-grass mixtures often provide more durable soil cover.
Green manure	±	Legume green manures can improve organic matter inputs, nitrogen availability, bulk density, and soil structure, potentially reducing erosion risk indirectly, although direct runoff or soil-loss evidence is limited.

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Table 2 (continued)

Erosion control driver	Effects	Reason
Temporary legume-based intercropping	±	Legume intercropping can reduce runoff and soil loss by increasing canopy cover, root inputs, organic matter, and aggregation; however, effectiveness depends on rainfall regime, biomass production, tillage intensity, and crop combination.
Crop management	±	Integrated legume-based systems can enhance soil cover, structure, infiltration, and erosion control, but their effectiveness depends on climate, species choice, residue management, soil disturbance, and compatibility with cereals, grasses, mulch, or conservation practices.
Overall synthesis		Legumes contribute to erosion control through combined canopy, rhizospheric, structural, and management effects. Their effectiveness is context-dependent and is often strongest when legume functions are combined with persistent cover from grasses or cereals and conservation-oriented management.

China, in a transition zone between northern subtropical and warm temperate climates, Li et al. (2023) investigated various cover crop systems in cherry orchards, including legume alfalfa (*Medicago sativa* L.), tall fescue (*Lolium arundinaceum* Schreb.), and legume-gramineae mixtures (alfalfa-tall fescue). Alfalfa and tall fescue produced the strongest reductions in sediment loss, approximately 53% and 52%, respectively, whereas the mixture reduced sediment loss by about 43%, all compared with plots without vegetation cover. However, when runoff was considered, the alfalfa-tall fescue mixture performed best, followed by alfalfa and tall fescue. This mixture reduced runoff by about 40% compared with natural grass cover. Thus, in this field experiment, mixed legume-grass systems may provide stronger overall water-loss control even when they are not the most effective treatment for sediment loss alone. Hu et al. (2023) compared artificial grasslands established with perennial legumes such as alfalfa and common sainfoin to natural grasslands on sloping farmland in the Loess Plateau. Artificial grasslands had lower canopy coverage (30–40%) than natural grasslands (83–87%) and showed weaker erosion control, with erosion rates around 0.20 t ha⁻¹ compared with 0.10 t ha⁻¹ under natural grassland. This suggests that the erosion-control performance of perennial legume-based artificial grasslands may be constrained by lower canopy cover and management disturbance, including mowing, harvesting, and periodic tillage, rather than by legume identity alone (Hu et al., 2023).

Overall, canopy effectiveness in legume-based systems is highly species- and context-dependent, being influenced by phenology, climate, and management intensity. While several legumes achieve rapid canopy closure and high surface cover, their performance may be constrained under less favourable conditions. A process-level synthesis of canopy-related responses is provided in Table 3.

3.3. Rainfall erosivity related to legume crop cycles and high R periods

The R-parameter, defined as the rainfall and runoff factor, estimates the erosive potential of rainfall by combining the effects of raindrop impact and runoff associated with precipitation intensity (Wischmeier and Smith, 1978). Higher R values result in greater soil particle detachment and transport. In Europe, the average annual erosivity is approximately 722 MJ mm ha⁻¹ h⁻¹ year⁻¹, with notably higher values recorded in Mediterranean and Alpine regions (>1000 MJ mm ha⁻¹ h⁻¹ year⁻¹) and lower values (<500 MJ mm ha⁻¹ h⁻¹ year⁻¹ in northern European regions (Panagos et al., 2015a).

The erosive potential is strongly influenced by rainfall intensity and storm events, as evidenced by the high erosivity in countries such as Italy, Spain, and Greece, where quickly and intense rainfall events pose a

Table 3

Selected quantitative evidence on legume-based systems and erosion-related processes.

The table reports representative quantitative examples supporting the Discussion, with emphasis on C-factor values, soil cover, runoff, soil and sediment loss, infiltration, hydraulic conductivity, root reinforcement, and aggregation. Values are presented as context-specific observations rather than pooled effect sizes, and illustrate how legume effects vary with species, management, reference treatment, and environmental conditions. The full dataset of extracted quantitative evidence is provided in Supplementary Table S1.

Process	Key quantitative evidence	Context/interpretation	Reference
Surface protection/C factor	Ccrop = 0.32 for dried pulses/protein crops; 0.21–0.23 for pure legume covers; 0.06–0.015 for dense legume-grass mixtures	Legume covers reduce C relative to bare soil, but dense legume-grass mixtures provide stronger surface protection than pure legume stands	(Novara et al., 2011; Panagos et al., 2015b)
Canopy development	Green Area Index development rate: 0.091 d ⁻¹ in faba bean, 0.061 d ⁻¹ in field pea, 0.022 d ⁻¹ in chickpea	Canopy closure differs strongly among legume species, affecting early soil protection	Mwanamwenge et al. (1997)
Seasonal cover reliability	Common vetch soil coverage declined from 96% to 47% between contrasting years	Legume cover performance may be strongly affected by seasonal conditions	Ramírez-García et al. (2015)
Legume vs cereal cover	Vetch cover: 44.4%; barley cover: 61.7%. Ks: vetch 23.2 mm h ⁻¹ , barley 36.1 mm h ⁻¹ , fallow 14.0 mm h ⁻¹	Legumes may improve soil hydraulic properties relative to fallow, but cereals can outperform legumes in rapid cover and Ks	(Gabriel et al., 2021; García-González et al., 2018)
Annual crop-cycle protection	Soybean vegetation cover increased from 10% to 73%, reducing runoff by 8–75% and soil loss by 13–92%	Increasing crop cover during the growth cycle can strongly reduce runoff and sediment loss	Yao et al. (2025)
Mediterranean vineyard cover crop	Soil erosion declined from 3.36 to 0.29 Mg ha ⁻¹ h ⁻¹ ; runoff coefficient from 32.9% to 13.7%; sediment concentration from 18.5 to 3.8 g L ⁻¹	Vetch cover can markedly reduce runoff and soil loss in bare or newly planted vineyard systems	Rodrigo-Comino et al. (2020)
Olive orchard cover management	Extreme-event soil loss: 0.86 vs 11.8 t ha ⁻¹ ; mean annual soil loss: 0.32 vs 3.25 t ha ⁻¹ yr ⁻¹	Full-surface legume cover under no-tillage provided much stronger protection than conventional tillage	Zuazo et al. (2020)
Legume-based intercropping	Runoff reduction: 17–43%; soil-loss reduction: 24–70%	Intercropping can reduce runoff and soil loss, although the magnitude varies with crop combination and context	Kumawat et al. (2022)

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Table 3 (continued)

Process	Key quantitative evidence	Context/interpretation	Reference
Living mulch/ maize intercropping	Chinese milk vetch reduced annual runoff by 61.2–70.0% and annual soil loss by 80.6–84.0%	Suitably selected legume living mulches can reduce erosion while maintaining crop-system function	Liu et al. (2025)
Root-derived infiltration	Alfalfa increased water penetration from 63 to 96 cm and steady infiltration by 31.80%	Deep-rooted perennial legumes can improve preferential flow and infiltration through root-derived channels	Guo et al. (2019)
Alfalfa cover under runoff scouring	Average infiltration increased from 0.220 to 0.546 mm min ⁻¹ ; runoff coefficient decreased by 28.3%; soil loss rate by ≈ 78%	Established alfalfa cover can improve infiltration and reduce runoff-driven sediment transport	Wu et al. (2011)
Root reinforce ment and aggregation	Root cohesion: 10.2 ± 3.4 kPa in birdsfoot trefoil, 7.6 ± 2.8 kPa in red clover, 7.8 ± 2.3 kPa in alfalfa; water-stable aggregates increased by 30.6–49.97% under perennial legumes	Perennial legumes can improve K-related soil resistance through root reinforcement and aggregation	(Comino et al., 2010; Guo et al., 2010)
Structural management in alfalfa	Open-ridging reduced runoff by 17.9–37.7% and sediment by 46.4–75.5%; tied-ridging reduced runoff by 22.3–55.5% and sediment by 62.1–87.6%	Management configuration can amplify erosion mitigation in alfalfa-based systems	Zhao et al. (2024)

greater erosion risk compared to wetter regions with more evenly distributed rainfall (Panagos et al., 2015a). In Italy, the average annual rainfall erosivity is estimated at 1351 MJ mm ha⁻¹ h⁻¹ year⁻¹, substantially higher than the European average. The summer months account for only 21% of rainfall events, yet they account for over 37% of annual erosion, with September being the month with the highest erosion rates and February the month with the lowest. These findings suggest that erosion patterns follow a Mediterranean pattern (Fig. 4) (Borrelli et al., 2016). Further regional evidence from central Italy reinforces this pattern (Fig. 5) (Todisco et al., 2025). This was clearly illustrated by Zuazo et al. (2020) in Mediterranean rainfed olive orchards on 20% slopes, where an extreme rainfall event caused 11.8 t ha⁻¹ of soil loss under conventional tillage, but only 0.86 t ha⁻¹ under a no-tillage treatment with legume cover over the entire soil surface. This result highlights the role of continuous full-surface cover under no-tillage, including legume-based cover, in buffering short-duration, high-intensity storms. Further, Baiamonte et al. (2019) compared vineyards managed under conventional practices (CP), involving frequent tillage and bare soil, with those under Best Management Practices (BMP), which included winter cover crops such as faba bean. The application of the RUSLE model at seasonal and event scales revealed that erosion risk increases when peaks in rainfall erosivity coincide with insufficient soil cover, particularly in September and February. BMP adoption resulted in an average erosion reduction of

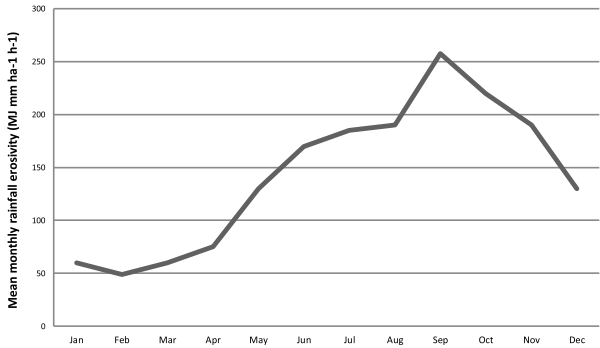


Fig. 4. Mean monthly rainfall erosivity in Italy (MJ mm ha⁻¹ h⁻¹), based on Borrelli et al. (2016).

3.2 t ha⁻¹ year⁻¹, with a maximum of 13.2 t ha⁻¹ recorded in a single year (Baiamonte et al., 2019). However, Baiamonte et al. (2019)'s study does not isolate the effects of legumes from those of general vegetative cover, which limits the ability to attribute the erosion reduction specifically to the presence of leguminous species. In a rainfall-simulation field experiment on soybean grown on gently sloping farmland, Yao et al. (2025) showed that runoff and sediment rates decreased from seeding to final flowering or grain formation and then increased again toward ripening. A more controlled small-plot laboratory study on red clover (*Trifolium pratense* L.) showed a consistent phenological pattern, with the flowering stage providing the strongest protection against runoff and especially soil loss (Naderi Marangalo et al., 2025). Yao et al. and Naderi Marangalo et al. indicates that the erosion-control performance of legumes is not constant throughout the crop cycle, but varies with phenological development.

Accordingly, the erosion-mitigating value of legume crops is strongly influenced by the duration and timing of their soil cover period. Murphy-Bokern et al. (2017) highlight the significant variability in leguminous plant crop cycles, allowing their integration across different seasons and production systems. Some species, such as faba bean, field pea, common and hairy vetch (*Vicia villosa* Roth.), lupin and lentil (*Lens culinaris* Medik.) and some clovers can be sown in both autumn and spring. Additionally, in temperate region, alfalfa, a perennial leguminous plant, can be established in either autumn or spring, offering stable and continuous cover for multiple seasons. Such diversification enables leguminous crops to have several agronomic roles, such as primary crops, cover crops, or components of crop mixtures, thus providing soil protection throughout different periods of the year (Murphy-Bokern et al., 2017).

Thus, legume species and management strategies should be selected to ensure effective soil cover during periods when erosive rainfall is more likely to coincide with vulnerable soil conditions. For the late

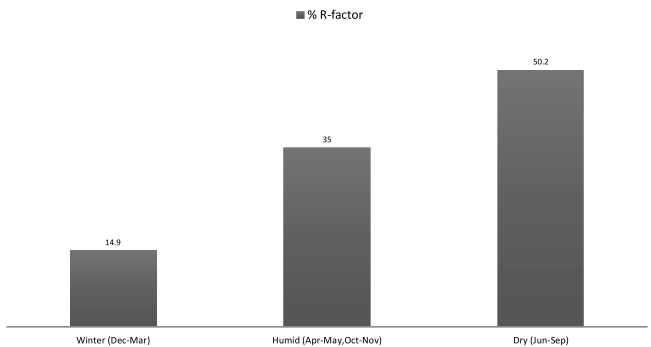


Fig. 5. Percentage distribution of mean annual rainfall erosivity (R) by four-month period in Umbria, central Italy, based on Todisco et al. (2025).

winter to spring period, autumn-sown species such as common vetch, hairy vetch, field pea, crimson clover and chickpea may contribute to soil cover, provided that establishment is early and sufficiently dense (Iglesias and Lloveras, 1998; Murphy-Bokern et al., 2017). Conversely, for the summer and autumn months (July–October), spring-sown species like lupin, field pea, and lentil may provide effective soil coverage during localized erosive events. However, they are effective only when canopy development is sufficiently rapid to precede erosive events. Soybean is a warm-season legume whose planting period and phenological development depend strongly on latitude, temperature, photoperiod, and cropping-system position (Burton, 1997). However, its early-season soil protection may be limited where wider row spacing and mechanical weed control delay full canopy closure (Murphy-Bokern et al., 2017). Perennial leguminous species such as alfalfa and white clover may provide longer-lasting soil cover once established, although their protective effect still depends on establishment success, persistence, and management (Murphy-Bokern et al., 2017).

4. Rhizosphere

4.1. Legume effects on K-related soil properties

The K factor represents soil erodibility, that is, the inherent susceptibility of soil to detachment and transport by raindrop impact and surface runoff. In the USLE/RUSLE framework, it is defined under standardized bare-soil plot conditions, with fixed slope length and steepness and without conservation practices, so that differences in estimated soil loss can be attributed mainly to soil properties (Ashiagbor et al., 2013; Wischmeier and Smith, 1978). The K factor represents the soil's inherent vulnerability to erosion by rainfall and runoff. K-factor is influenced by several soil properties, including soil particle size distribution (i.e., the proportion of sand, silt, and clay), organic matter content (which increases soil cohesion and aggregate stability), soil structure (the spatial arrangement of aggregates, influencing soil quality and resistance to detachment), and permeability (the soil's capacity to transmit water, affecting infiltration and runoff generation) (Brady and Weil, 2016; Wischmeier and Smith, 1978). Higher K values indicate greater susceptibility to detachment and transport. Leguminous plants may influence soil properties correlated to soil erodibility by improving multiple soil properties that directly impact soil erodibility. The capacity of legume root systems to reduce erosion is supported by both physical reinforcement and rhizospheric effects. In situ shear tests showed that bird's-foot trefoil (*Lotus corniculatus* L.), red clover and alfalfa roots increased additional soil strength provided by plant roots (root cohesion) by 10.2 ± 3.4 ; 7.6 ± 2.8 and 7.8 ± 2.3 kPa respectively (Comino et al., 2010). Hinsinger et al. (2009) reviewed evidence that rhizosphere microbial activity and root-derived mucilage can influence soil structure and aggregation. Additional evidence from the northern region of the semi-arid Loess Plateau in China showed that converting cropland to perennial grassland cover with alfalfa and sweet clover (*Melilotus officinalis* L.), significantly improved soil physical properties associated with erodibility. The authors showed that after five years, soil bulk density declined and water-stable aggregates rose by 49.97% under sweet clover and by 30.6% under alfalfa (Guo et al., 2010). Khan et al. (2022) showed that continuous alfalfa had the highest content of water-stable aggregates in the upper 20 cm of soil, together with greater root biomass and the highest mean and geometric aggregate diameters, while soil organic carbon and total nitrogen were positively associated with aggregate stability. Thus, long-term alfalfa cultivation improved several structural indicators associated with lower soil erodibility, especially in the upper soil layers (Khan et al., 2022). A recent 16-month outdoor pot experiment in northeast China simulating progressive Mollisol erosion further confirmed that perennial legumes can improve structural indicators associated with lower soil erodibility. Compared with bare soil, alfalfa and milkvetch (*Astragalus adsurgens* Pall.) increased mean weight diameter by 65–113% and reduced the potential

structural deformation index by 10–19%, particularly under no to moderate erosion, indicating a marked aggregate-stabilizing effect of Fabaceae under the experimental conditions (Zhang et al., 2026). In Mediterranean olive groves, common vetch cover under no-tillage increased macro-porosity, soil organic carbon, and cation exchange capacity compared with conventional tillage without plant cover, and was associated with lower runoff and soil loss (Rodríguez et al., 2022). These effects suggest reduced erosion susceptibility, although Rodríguez et al. (2022) did not directly quantify soil erodibility or the RUSLE K factor. Blanco-Canqui and Ruis (2020) reported that cover crops including red fescue, alfalfa and bird's-foot trefoil increased average macropore diameter from 0.78 mm to 1.14 mm and mesopores from 0.14 mm to 0.25 mm, compared to fallow or conventional treatments.

Legume-based mixtures can influence soil surface hydraulic conductivity (mm h^{-1}) that is the capacity of the soil surface to transmit infiltrating water. Higher hydraulic conductivity promotes infiltration and reduces surface runoff, thereby decreasing the potential for soil detachment and transport by overland flow. In a semi-arid reclaimed mine area of the northern Loess Plateau a 75 min infiltration plot experiment showed that the combination of alfalfa and smooth brome (*Bromus inermis* Leyss.) produced the highest average infiltration rate ($26.85 \pm 2.51 \text{ mm h}^{-1}$), slightly exceeding that of alfalfa alone ($26.6 \pm 1.88 \text{ mm h}^{-1}$) and smooth brome alone ($25.81 \pm 1.98 \text{ mm h}^{-1}$); during the first 15 min, which reflect early-stage infiltration, the mixture achieved $49.93 \pm 4.56 \text{ mm h}^{-1}$, compared to $46.6 \pm 2.33 \text{ mm h}^{-1}$ under smooth brome and $57.33 \pm 7.53 \text{ mm h}^{-1}$ with alfalfa (Wu et al., 2016). Thus, combination of fibrous roots and taprooted species can provide complementary functional traits enhancing infiltration, although early-stage infiltration was highest under alfalfa alone.

Legume-associated rhizobia and AMF may indirectly affect soil structure through nutrient cycling, microbial activity, and rhizosphere binding agents. For instance, Kabalan et al. (2022) observed that purple vetch (*Vicia Benghalensis* L.) retained the highest amount of glomalin-related soil proteins among the tested cover crops, indicating a possible role in soil particle binding and aggregate stability, although unclear contribution. Root-derived compounds may also contribute indirectly to soil structural stability. Mucilage and microbial exopolymers can promote soil aggregation and root-soil contact, while organic acids mainly influence nutrient mobilization and rhizosphere chemistry rather than resistance to erosion directly (Bais et al., 2004, 2006; Haichar et al., 2014; Jones et al., 2009).

Overall, legume-based systems can improve several soil properties associated with lower erodibility, including root cohesion, aggregate stability, pore structure, and infiltration capacity (Comino et al., 2010; Khan et al., 2022; Rodríguez et al., 2022; Wu et al., 2016; Zhang et al., 2026). The quantitative ranges reported in Table 3 are context-specific responses rather than pooled estimates, because they derive from different experimental designs, reference treatments, species, and soil-climate conditions.

4.2. Mechanical properties

Herbaceous legumes can contribute to soil stabilization through root-mediated mechanical effects, particularly by increasing root-soil cohesion, anchorage, and aggregate stability. However, these effects are strongly species-dependent and vary with root architecture, soil type, and management. Comino et al. (2010) demonstrated that common bird's-foot trefoil, had a high root density of approximately 5500 roots m^{-2} and produced the highest additional root cohesion among the tested species (10.2 ± 3.4 kPa), indicating a strong capacity to reinforce the soil-root matrix. Xu et al. (2007) reinforce this, highlighting that alfalfa and milkvetch exhibit highly developed root systems within the top 60 cm of soil, this suggests a potential contribution to near-surface soil reinforcement. Cao et al. (2025) provided further evidence, from a controlled pot experiment followed by laboratory mechanical testing and slope-stability simulation in the Fangta watershed, a temperate

semi-arid loess hilly-gully area in Shaanxi Province, China. The authors used silty loam soil collected from the 0–30 cm layer and compared tap-rooted alfalfa with fibrous-rooted perennial ryegrass (*Lolium perenne* L.), a non-legume reference species commonly used for soil and water conservation in the region. Their results show that root effects were expressed mainly through changes in cohesion rather than in internal friction angle (that quantifies the frictional resistance between soil particles during shear), with maximum cohesion increases of 34.4% for alfalfa and 80.57% for perennial ryegrass relative to the control. A broader trait-based assessment by Wang et al. (2026) from the Zhi-fanggou small watershed, further showed that the erosion-reducing effect of herbaceous root systems is better explained by an integrated combination of root traits. Across eight representative herbaceous species from the Loess Plateau with a transition zone between semi-humid and semi-arid conditions, the composite root system parameter, based on root surface area density, root mass density, root cohesion, and topology index, was 1.41 times higher in fibrous-rooted species including wild rye (*Leymus secalinus* (Georgi) Tzvelev), Bunge needlegrass (*Stipa bungeana* Trin.), and yellow bluestem (*Bothriochloa ischaemum* (L.) Keng), than in tap-rooted species such as including alfalfa and milk-vetch. Relative soil detachment capacity was 14% lower in fibrous-rooted species and declined exponentially as the integrated root parameter increased. Yu et al. (2016) in a field experiment conducted in Austria, found that legumes such as sweet clover and grass pea (*Lathyrus sativus* L.) exhibited markedly higher saturated hydraulic conductivity ($K_s = 25.7$ and 24.8 cm h^{-1} , respectively) compared to grasses like rye (16.6 cm h^{-1}). Across species K_s was also higher in the crop row than in the inter-row (23.22 vs 10.08 cm h^{-1}). This pattern was associated with their larger root diameters (up to 0.49 mm). However, the study also highlighted notable variability among legume species, with berseem clover (*Trifolium alexandrinum* L.) showing a lower K_s (15.4 cm h^{-1}) than sweet clover and grass pea. K_s was generally higher in the row rather than in the inter-row, indicating both species identity and root spatial distribution influenced soil hydraulic behaviour. In contrast, Henry et al. (2009) found that root systems of annual crops such as soybean and common bean, had minimal direct effects on soil erosion and runoff if this is characterized by low root length densities ($<4.0 \text{ cm cm}^{-3}$). Thus, direct root-mediated erosion control is more likely in high-density root systems, such as grass systems, than in annual dicotyledonous row crops with relatively sparse roots and low sowing density.

Root legacy effects after plant death are also relevant. According to Qu et al. (2025), the decomposition of roots forms stable and persistent macro-pores, resulting in a 20.6 fold increase in effective macro-porosity ($>3 \text{ mm}$) and a steady-state infiltration rate up to 6.95 mm/min , 5.7 times higher than in natural recovery grassland with dense live roots. This evidence supports a root specific effect mechanism, which it is not specific to legumes. Qu et al. (2025) also showed that live roots may occupy pre-existing macro-pores and reduce pore connectivity. In natural recovery grassland, effective macro-porosity $>3 \text{ mm}$ declined by 93.67% at 20 cm depth, and infiltration rate decreased by 74.31%, indicating that dense live root systems can either promote or obstruct preferential flow depending on depth and pore connectivity. Likewise, in semi-arid artificial grassland, Liu et al. (2020) showed that infiltration decreased with increasing root length density in the 0–10 and 20–30 cm layers, as well as with higher root surface area at 10–20 cm. Thus, dense fibrous root systems, such as those of smooth brome, tended to obstruct flow pathways and reduce infiltration (measured SIR: 52.69 mm h^{-1}), whereas alfalfa showed higher infiltration (measured SIR: 66.69 mm h^{-1}). Together these studies help explain how root architecture can influence soil hydraulic behaviour and, indirectly, erosion-related processes.

In terms of erosion control, Sharma et al. (2018) and De Baets et al. (2008) highlight that fibrous root systems are generally more effective than taproot systems in reducing detachment, particularly because they provide greater near-surface root density and root-soil contact area. De

Baets et al. (2008) in a lab experiment, showed that grass roots increased soil cohesion by $3.41 \pm 2.36 \text{ kPa}$, compared with $1.66 \pm 1.66 \text{ kPa}$ for carrot taproots, indicating a stronger near-surface reinforcement effect of fibrous root systems in the tested silt loam topsoil. This superior performance is attributed to the greater root-soil contact area of fibrous roots, which enhances the binding of soil particles and resistance to concentrate-flow detachment.

These findings suggest that dense fibrous roots can strongly reinforce the near-surface soil matrix by binding particles and reducing detachment, although very dense live root systems may also obstruct preferential flow pathways and limit infiltration under some soil and rooting conditions. Conversely, coarse-rooted or taprooted species may favour vertical water movement by increasing coarse macro-pores or hydraulic conductivity, but their contribution to near-surface cohesion is often less pronounced than that of dense fibrous root systems. The mechanical role of herbaceous root systems in erosion control should therefore be interpreted as the outcome of complementary functions: near-surface stabilization against detachment, improvement of soil cohesion, and modification of infiltration pathways. For legumes, these effects depend strongly on species, root architecture, stand persistence, and management. Therefore, improvements in root reinforcement, infiltration, or soil structure should not be assumed to translate automatically into lower soil loss unless runoff, or soil-loss outcomes are directly measured (Cao et al., 2025; Hu et al., 2023; Van Eerd et al., 2023; Wang et al., 2026).

4.3. Water infiltration and runoff dynamics

Infiltration is a dynamic process which regulates when rainfall trigger soil surface runoff. During the early phase of a rainfall event, infiltration capacity may exceed rainfall intensity limiting runoff generation. As rainfall continues, soil wetting, surface sealing, aggregate breakdown, and pore clogging can reduce infiltration capacity increasing runoff volume and the transport capacity of overland flow (Brady and Weil, 2016; Todisco et al., 2023).

Perennial and deep-rooted vegetation, including legumes, can improve infiltration dynamics and soil structure through the development of root derived macro-pores and preferential flow pathways (Cui et al., 2019; McCallum et al., 2004). In China Lei Guo et al. (2019) demonstrated that alfalfa roots substantially improved infiltration, allowing water to reach depths of 96 cm compared to 63 cm in bare soil, due to preferential channels created by its decayed taproots. Further, the steady infiltration rates increased by 31.80% relative to bare soil. Complementary evidence from runoff-scouring experiments on 20° loess slopes showed that established alfalfa cover increased the average infiltration rate from 0.220 to $0.546 \text{ mm min}^{-1}$, while reducing the runoff coefficient by 28.3% and the soil loss rate by approximately 78% compared with bare soil (Wu et al., 2011). Because this experiment was based on overland-flow scouring rather than rainfall interception, its results mainly support the role of alfalfa cover in increasing hydraulic resistance, dissipating flow energy, and reducing sediment transport. This response is consistent with trait-based evidence showing that sediment retention and runoff control in herbaceous vegetation are strongly promoted by aboveground traits such as stem density, stem diameter, projected stem area, and leaf area, which increase hydraulic resistance and dissipate flow energy along the slope (Kervroëdan et al., 2021; Wu et al., 2011).

Qu et al. (2025) in a study conducted in China with temperate semi-humid monsoon conditions showed that both live and dead roots can contribute to macropore formation, but long and continuous biogenic macro-pores formed after root death had a stronger effect on soil water infiltration than living roots. In the same study, effective macro-porosity greater than 3 mm increased markedly after root decay, whereas live roots could also reoccupy or block existing macro-pores in deeper soil layers, reducing their hydraulic effectiveness. Other evidence from Loess Plateau with semi-arid conditions, was reported in

systems initially dominated by alfalfa, where subsequent root decomposition left preferential channels that facilitated surface-layer and deeper water movement (Cui et al., 2022; Wu et al., 2019). Fang et al. (2024) demonstrated that root decomposition enhanced soil water infiltration in both gramineous and leguminous forages. After 450 days of decomposition, Dahurian lespedeza (*Lepedeza davurica* (Lax.) Schindl.) increased initial, steady-state, and average infiltration rates by 38.3%, 267.7%, and 137.1%, respectively, compared with 18.4%, 106.5%, and 61.6% with the Yellow Bluestems. These findings suggest that decomposed legume roots may promote infiltration through changes in root volume, root length, soil porosity, and aggregate formation, although this response should not be generalized to all legume systems because it depends on species traits, decomposition stage, soil conditions, and management history. Increased infiltration under legume-based systems can also be linked to enhanced earthworm activity. Marley et al. (2024) under temperate maritime conditions in Wales (UK), reported significantly greater earthworm biomass under white clover compared with perennial ryegrass ($\alpha < 0.05$), in a stony well-drained loam, suggesting that earthworm burrowing activity may have contributed to the formation of preferential flow pathways that increased surface water infiltration. However, this mechanism might not be fully legume specific.

Blanco-Canqui and Ruis (2020) found that most cover crop treatments increased infiltration, with reported increases of up to 445% when common vetch was combined with rye (*Secale cereale* L.) and triticale ($\times$ *Triticosecale* Wittmack). In a 10-year irrigated rotation trial in central Spain, common vetch increased soil saturated hydraulic conductivity (Ks) only slightly compared to fallow (23.2 vs 14.0 mm h⁻¹), whereas barley markedly improved Ks, reaching 36.1 mm h⁻¹ (García-González et al., 2018). A similar caution emerges from an experiment in south-west Queensland on a relatively free-draining, low-clay soil, where clay content increased from 12.4% in the 0–0.1 m layer to 24.3% at 1.2–1.5 m depth. In this study, cumulative infiltration after 100 mm of simulated rainfall was higher following continuous wheat under zero tillage than after chickpea under zero tillage (52 vs 39 mm). Short-term pasture leys based on alfalfa, snail medic (*Medicago scutellata* (L.) Mill.) and barrel medic (*Medicago truncatula* Gaertn.) also did not clearly improve infiltration characteristics, with mean cumulative infiltration of 42 mm after an alfalfa-snail medic-barrel medic ley and 45 mm after a grass-alfalfa-snail medic-barrel medic ley, compared with 52–63 mm after continuous wheat treatments. These results indicate that, under the soil and management conditions of this experiment, legume-based systems did not outperform cereal-based references in terms of rainfall infiltration (Thomas et al., 2008). In contrast, when legumes were included in multispecies cover crop mixtures, Haruna et al. (2022) reported significantly greater cumulative infiltration and Ks in soils managed with a multispecies cover crop mixture including legumes compared with plots without cover crops. More broadly, a synthesis of the literature reported that cover crops generally enhance soil infiltration, although the magnitude of the effect varies widely depending on soil properties, climate, and management practices (Çerçioğlu et al., 2025).

Under simulated rainfall, alfalfa reduced runoff relative to bare soil, but perennial ryegrass and spring wheat were more effective: mean runoff intensity decreased from 0.664 mm min⁻¹ under bare soil to 0.500 mm min⁻¹ under purple medic, 0.437 mm min⁻¹ under spring wheat, and 0.300 mm min⁻¹ under perennial ryegrass (Huang et al., 2013). Liu et al. (2019) compared four typical grasslands on the Loess Plateau white clover, alfalfa, dahurian wheatgrass (*Elymus dahuricus* Turcz. Ex Griseb.), and smooth brome under two-year simulated rainfall experiments on 20° slopes. The study showed that all grasslands reduced soil loss relative to bare soil, but the strongest sediment reduction and runoff-maintenance performance was observed in the two gramineous treatments. In the second year, sediment reduction benefits reached 98.79% under dahurian wheatgrass and 98.07% under smooth brome, whereas alfalfa showed a substantially lower sediment reduction

response. These results indicate that, in this experiment, grasses were more effective than legumes in combining sediment reduction with runoff maintenance, mainly due to dense fibrous roots and higher biological soil crust coverage. A phenological rainfall-simulation study on red clover further showed that runoff regulation varies across crop development stages. On 20% slopes under 50 mm h⁻¹ simulated rainfall, runoff volume was reduced by 21.5%, 25.9%, 39.5%, 33.9%, and 17.0% at the germination, greening, flowering, live-root, and dead-root stages, respectively, relative to bare soil. Peak runoff was also reduced by 24.4%, 29.7%, 39.7%, 29.0%, and 17.0% across the same stages, with the strongest response occurring at flowering (Naderi Marangalo et al., 2025).

Kocira et al. (2020) summarized evidence that legume cover crops such as hairy vetch increased water-stable aggregates by 9–17%, and mixtures including crimson clover further enhanced aggregate formation compared with soils without cover crops. In Mediterranean olive groves, Sastre et al. (2018) show that common sainfoin, a tap-rooted legume, developed a higher root density than annual barley in the upper 10 cm of soil (≈ 8 kg m⁻³) and maintained root presence in deeper layers. Sainfoin increased soil moisture at 30 cm depth, reaching 0.17 m³ m⁻³ in winter about 20% higher than the control treatment. However, steady-state infiltration did not differ significantly among treatments, while the perennial grass stiff brome (*Brachypodium distachyon* (L.) P. Beauv.) showed the highest water content across all seasons (up to 0.20 m³ m⁻³ at 30 cm depth in winter) and improved aggregate stability and SOC.

Henry et al. (2009) argued that, for annual legumes (e.g. common beans and soybeans) canopy coverage rather than root systems plays a more significant role in mitigating erosion. In their rainfall simulation trials, Henry et al. (2009) expressed runoff as the cumulative volume of surface water converted to an equivalent depth (mm) over the plot area. Under intact canopies, runoff ranged from 1.3 to 7.5 mm, whereas removal of plant shoots increased runoff to 7.0–16.8 mm this metric reflects the equivalent water height over the plot surface, with lower values indicating reduced runoff generation under intact canopies. Consistent with this pattern, Thomas et al. (2008) reported on a relatively free-draining, low-clay soil, in south-west Queensland with different type of canopy cover, leguminous and non-legumes, that the mean cumulative infiltration after 100 mm of simulated rainfall increased from 35 mm under bare soil to 42 mm under existing cover and 69 mm under full cover. This interpretation is consistent with Naderi Marangalo et al. (2025), who showed that the effectiveness of red clover in reducing runoff and soil loss varied across phenological stages, highlighting the importance of plant development for erosion control. These results indicate that runoff regulation became stronger as canopy and root systems developed.

In a comparative field study on the Loess Plateau in China, Huang et al. (2017) compared the infiltration capacity of alfalfa, wheat grass (*Agropyron cristatum* (L.) Gaertn.), Korshinsk pea shrub (*Caragana korshinskii* Kom.), and stipa capillata (*Stipa capillata* L.). Legume grasslands exhibited significantly higher mean infiltration rates, with alfalfa reaching 37.5 mm h⁻¹ compared with 24.3–27.3 mm h⁻¹ for grasses ($\alpha < 0.05$). This response was associated with greater below-ground biomass in the 10–30 cm layer, which was one of the main positive predictors of infiltration rate in that study. By contrast, artificial grasslands on reclaimed mine soils generally improved infiltration capacity relative to bare soil, with particularly favourable responses in legume-shrub and legume-Poaceae mixtures (Wu et al., 2016). However, the response was mainly associated with belowground biomass, soil water content, and capillary and non-capillary porosity, rather than with legume identity alone. A broader synthesis of arid and semi-arid grasslands provided by Jia et al. (2024), further suggests that these contrasting responses arise because roots influence infiltration through multiple and partly opposing pathways. For example, higher root length density, root surface area density, and root biomass can enhance infiltration by increasing non-capillary porosity, root channels, and

aggregate formation. However, very dense live root systems or shallow fibrous/rhizomatous root mats may also occupy or block existing pores and reduce water movement. Conversely, root decay can create persistent macro-pores and preferential flow pathways; in alfalfa plots, Jia et al. (2024) reported that decayed roots increased initial and steady infiltration rates by 27.70% and 31.80%, respectively, compared with bare field conditions. Their quantitative synthesis indicated that non-capillary porosity accounted for more than 50% of the positive root effect on infiltration, whereas root-induced increases in soil organic matter could offset about 25% of this benefit.

Overall, legume-based systems can contribute to infiltration and runoff regulation. However, their effects are strongly mediated by species identity, root architecture, phenological stage, root decomposition, soil properties, and comparator type. Perennial or tap-rooted legumes may enhance preferential flow through root-derived macro-pores, particularly after root decay, whereas annual legumes often influence short-term runoff mainly through canopy development and surface cover. Thus, legumes are not uniformly superior to grasses and cereals, especially when fibrous roots, persistent cover, or mixed-species systems provide stronger surface protection and hydraulic resistance. Indeed, infiltration and runoff response are a result of the combination processes mediated by pedoclimatic conditions interactions rather than legume characteristics. A consolidated overview of infiltration-related responses and reported ranges is provided in Table 3.

4.4. Symbiosis

Leguminous species interact with a broad range of soil microorganisms, among which rhizobia play a central role in improving plant nutrition, nodulation, and soil fertility under cover crop and green manure systems. Sun and Shahrajabian (2022) highlighted that these symbiotic interactions enhance nutrient cycling and root development, providing a biological basis for subsequent improvements in soil structure and related erosion-mitigating functions. AMF significantly enhance soil fertility through their symbiotic interactions with plants, including leguminous species (Jones et al., 2009; Kabalan et al., 2022). Kabalan et al. (2022) reported that among the tested cover crops, faba bean exhibited the highest AMF colonization frequency.

Haynes and Beare (1997) in a glasshouse pot experiment, demonstrated that legumes such as lupin and white clover achieve unexpectedly high aggregate stability despite having shorter root systems. Lupin showed the highest aggregate stability with a mean weight diameter (MWD) of 0.49 mm, significantly higher than winter wheat (0.30 mm), even though its root length density was about 60% lower (5.1 cm cm^{-3} vs. 13 cm cm^{-3}). White clover matched the aggregate stability of Italian ryegrass (*Lolium multiflorum* Lam.) (MWD $\approx 0.43 \text{ mm}$) despite having the lowest root length density among all species tested. This superiority is largely due to the symbiotic relationships that legumes establish, fostering distinct rhizosphere microbial communities and greatly enhancing fungal hyphal length measured at 1224 mg^{-1} under lupin, about four times greater than the 310 mg^{-1} observed under winter wheat. Furthermore, Kabalan et al. (2022) discuss that AMF enhance drought resistance by improving the plant's capacity to access water. In summary, symbioses between leguminous plants, rhizobia, and AMF play an essential role in improving soil fertility and enhancing soil structural stability.

4.5. Chemical properties

Legume cover crop can increase organic matter and nitrogen inputs and, through their effects on soil structure and biological activity, may improve aggregate stability and structural resilience, with potential benefits for soil erodibility and long-term soil conservation (Blanco-Canqui and Ruis, 2020; Kocira et al., 2020). Root exudates released by plants, including legumes, consist of low-molecular-weight compounds such as organic acids (e.g., citric and malic), amino acids,

sugars, and secondary metabolites, as well as high-molecular-weight substances like mucilage. These compounds enhance microbial activity and regulate nutrient dynamics within the rhizosphere, contributing to soil fertility (Bais et al., 2006; Bertin et al., 2003; Haichar et al., 2014). In addition, mucilage and other root- or microbe-derived polysaccharides may affect soil structure and water relations in the rhizosphere, with potential implications for aggregate stability and root-soil contact (Hinsinger et al., 2009; Jones et al., 2009). Organic acids such as citrate and malate are also involved in nutrient mobilization, particularly phosphorus, through rhizosphere chemical interactions, although their erosion-related effects remain indirect and context-dependent (Jones et al., 2009).

Haichar et al. (2014), Bais et al. (2006, 2004) and C. Bertin et al. (2003) highlighted that root exudates mediate chemical communication in the rhizosphere, facilitating symbiotic relationships with beneficial microorganisms such as rhizobia and AMF. Through these interactions, root exudates can influence microbial community structure and nutrient dynamics in the rhizosphere, with potential implications for plant nutrition and soil functioning.

Additionally, legume crops have a stronger effect on soil aggregation than non-legume species. Hinsinger et al. (2009) discussed the role of legume rhizosphere microbial community in promoting soil structural stability, particularly saprophytic fungi associated with legume crops. Yakupoglu et al. (2011) reported annual forage legumes from the genera *Lathyrus* sp and *Vicia* sp markedly enhanced soil aggregate stability in a clay soil, increasing wet aggregate stability (WAS) up to 77.7% and reducing the dispersion ratio to as low as 6.4%. Round-seeded vetchling (*Lathyrus sphaericus* Retz.) and yellow vetch (*Vicia lutea* L.) were the most effective species. The magnitude of these improvements varied by species and soil depth, likely in relation to differences in rooting patterns. Sastre et al. (2018) highlighted that although slight improvements in soil quality and aggregate stability were observed after three years of cover crop management, longer period of continuous application are required to achieve significant soil restoration in a gypsiferous soil classified as Haplic Gypsisol.

5. Agronomical practices

5.1. Conservation tillage

Tillage practices, including conventional, reduced, and no-tillage systems, directly affect soil surface conditions by modifying residue cover, surface roughness, and soil structure throughout the crop cycle. These dynamic changes influence the soil's susceptibility to raindrop impact and surface runoff, thereby impacting erosion rates. Consequently, tillage is incorporated into the C factor of the USLE/RUSLE model, which accounts for the combined effects of crop cover and management practices on soil loss during the growing season (Wischmeier and Smith, 1978).

According to Bronick and Lal (2005) and Skaalsveen et al. (2019), conventional tillage can disrupt soil aggregates, expose organic matter to faster mineralization, and leave the soil surface more sensitive to raindrop impact and runoff, thereby increasing erosion susceptibility with limited residue cover. Reduced and no-till systems can promote soil organic carbon (SOC) accumulation and aggregate stability near the soil surface, particularly when residue cover and cover crops are maintained. In NW European studies reviewed by Skaalsveen et al., soil loss reductions under no-till ranged from about 71% in rainfall-simulation experiments to 90-100% in some modelled event-based scenarios. However, in the absence of a sufficient surface mulch or crop residue on the topsoil, no-till may promote surface sealing and topsoil compaction, reducing infiltration, increasing runoff, and, in some cases, exacerbating erosion rates (Giller et al., 2015; Skaalsveen et al., 2019).

Conservation practices also influence rhizosphere biological interactions. A global meta-analysis of 54 field studies showed that alternative tillage increased AMF colonization of cash crop roots by 27%

compared to conventional tillage, while no-till increased colonization by 30.3%. Cover cropping, particularly with legumes such as hairy vetch and clovers, further enhanced colonization (Bowles et al., 2017). These results suggest that combining no-till with legume cover crops can increase AMF colonization probably because of their high mycorrhizal dependency and contribution to soil inoculum potential. By promoting below-ground biological interactions, these practices may support ecological intensification and contribute indirectly to soil structure formation and biologically based resource use. However, their effects on erosion and nutrient losses should be interpreted as potential downstream outcomes rather than direct results measured by Bowles et al. (2017). Management effects on legume root development are also illustrated by field studies on faba bean. A three-year field study on vertisols in southern Spain showed no-till management significantly enhanced faba bean root development compared to conventional tillage (Muñoz-Romero et al., 2011). Root biomass was higher under no-till than under conventional tillage in two of the three growing seasons. Muñoz-Romero et al. (2011) attributed the greater root development under no-till to improved soil physical conditions and better soil water status. Root biomass was also positively associated with aboveground biomass and grain yield, although the magnitude of the no-till response varied among years (Muñoz-Romero et al., 2011).

5.1.1. Cover crops

Among the agronomic practices integrated within conservation tillage systems, the use of cover crops has attracted increasing attention for its potential role in reducing soil erosion and improving soil conditions. Cover crops (CCs) are non-harvested plants grown between main crops or during fallow periods, primarily to protect and improve the soil. Their use has a long-standing tradition in sustainable agriculture and is increasingly emphasized in conservation systems due to their capacity to limit erosion, increase soil organic matter, enhance biological activity, and regulate nutrient dynamics (Kocira et al., 2020).

The erosion-reducing effects of CCs are mainly linked to surface cover, which intercepts raindrop impact and reduces runoff velocity, and to longer-term improvements water infiltration, soil porosity and aggregate stability (Blanco-Canqui and Ruis, 2020).

Recent reviews show that cover crops can improve soil physical and hydraulic properties. Haruna et al. (2020) reported that cover crops reduced bulk density by approximately 4%, increased macro-pores by approximately 33%, and increased water infiltration by up to 629% compared with no-cover controls. The largest infiltration response was reported for winter annual legume and rapeseed cover crops, where infiltration increased by 416% and 629%, respectively, mainly due to root-derived macro-pores. The same review also reported soil-loss reductions of up to 96% in no-till soybean systems with winter cover crops in Missouri. However, these values represent upper-range responses rather than general expectations: Blanco-Canqui and Ruis (2020) found that cover crops increased infiltration in most studies but had negligible or inconsistent effects on bulk density, hydraulic conductivity, and water retention. Therefore, these values should be interpreted as context-specific ranges rather than as a single causal sequence. In addition to improving soil structure and hydraulic properties, CCs contribute to soil carbon sequestration and fertility through residue inputs and root activity (Blanco-Canqui et al., 2015). In New Mexico a semi-arid no-till silage systems, showed that crop mixtures including grasses, brassicas, and legumes, improved structural indicators linked to soil erodibility. Acharya et al. (2024) reported that, at 0–0.1 m depth, mean weight diameter and geometric mean diameter increased by 22–23% and 6%, respectively, while SOC in the 0.25–2, 0.053–0.25, and <0.053 mm aggregate fractions increased by 15–17%, 15–16%, and 13% relative to the no-cover control, indicating greater aggregate stabilization and organic matter protection under semi-arid erosion-prone conditions. However, aggregate-stability responses were inconsistent across years.

Leguminous cover crops contribute to soil conservation through a

combination of biological and physical effects. Legumes through fixing atmospheric nitrogen and with the provision of biomass residue to the soil can enhance microbial activity and improve soil structure, making them useful components of integrated management systems (Kocira et al., 2020; Scavo et al., 2022). According to Kocira et al. (2020), these crops can simultaneously support fertility restoration and reduce erosion risk, especially when integrated into conservation agriculture strategies. In a three-year rainfall simulation study on newly planted vineyards, common vetch reduced soil erosion from 3.36 to 0.29 Mg ha⁻¹ h⁻¹ and the runoff coefficient from 32.9% to 13.7%, while sediment concentration declined by nearly 80% (18.5 to 3.8 g L⁻¹). Plant cover delayed runoff initiation by a factor of 3.7, demonstrating the effectiveness of common vetch in reducing soil and water losses in Mediterranean sloping vineyards (Rodrigo-Comino et al., 2020). In Mediterranean rainfed olive orchards on 20% slopes, Zuazo et al. (2020) showed that the anti-erosive effect of legume-based cover depended strongly on how the cover was implemented. The no-tillage treatment with legume cover over the entire soil surface (LEC) produced the lowest mean soil loss among all individual treatments (0.32 t ha⁻¹ year⁻¹), compared with conventional tillage (3.25 t ha⁻¹ year⁻¹), legume strips under minimum tillage (2.15 t ha⁻¹ year⁻¹), legume strips under no-tillage (2.50 t ha⁻¹ year⁻¹), and even spontaneous vegetation over the whole soil surface (1.07 t ha⁻¹ year⁻¹). These results indicate that leguminous cover crops can provide strong erosion control. However, their effectiveness is maximized when they ensure continuous soil cover under minimal disturbance. Hartwig and Ammon (2002) further reported that perennial legume living mulches can provide very strong erosion control when combined with reduced soil disturbance. In maize grown on a 14% slope, no-till systems with birdsfoot trefoil or crown-vetch (*Securigera varia* (L.) Lassen) living mulches reduced water runoff by about 96% and soil loss by 99.6–99.9% relative to conventional tillage, highlighting the protective value of continuous legume ground cover under no-till conditions. In the western coastal region of Syria, winter legume-based cover systems significantly reduced soil loss, runoff, and eroded organic matter under natural rainstorms compared with both bare soil and wheat. At 8% slope, cumulative soil loss declined from 0.468 kg m⁻² in the control and 0.368 kg m⁻² under wheat to 0.170 kg m⁻² under hairy vetch and 0.154 kg m⁻² under bean, while at 20% slope bean remained the most effective treatment (0.489 kg m⁻² versus 0.769 in hairy vetch, 0.967 in wheat, and 1.145 in the control), supporting the protective role of legume cover under Mediterranean sloping conditions (Mohammed et al., 2021). In olive groves managed with subterranean clover as a cover crop, Hernández et al. (2005) reported significant improvements in soil water retention and storage capacity. After three years, soil field capacity increased from 4.69 to 5.79%, and available water rose from 3.34 to 4.25%, corresponding to a 27.2% enhancement in water-holding capacity. The subterranean clover treatment produced substantial soil cover, with bare soil declining from 52% in the first year to 22%, 16%, and 9% in the third, fourth, and fifth years, respectively. Hernández et al. (2005) described this cover as an effective protective mesh against spring storm erosion and winter surface runoff, although runoff and soil detachment were not quantified directly. These changes suggest improved soil water storage under subterranean clover cover. The protective cover may also have reduced surface sealing and runoff risk, but infiltration, hydraulic conductivity, and aggregate stability were not directly quantified in this study (Hernández et al., 2005). In semi-arid gypsiferous olive groves of central Spain, cover crops significantly reduced soil loss relative to tillage, but their performance differed according to cover type: annual legume covers reduced soil loss by about 40%, compared with 60% under barley and 80% under permanent stiff brome cover. In the same experimental system, three years of management showed that sainfoin produced only limited improvements in soil quality, with water-stable aggregates increasing from 32.3% in the tilled control to 37.0%, whereas stiff brome reached 44.3%. Likewise, soil organic carbon in the 0–10 cm layer was highest under stiff brome (13.7 Mg ha⁻¹) and lower under sainfoin

(8.8 Mg ha⁻¹), indicating that legume cover can contribute to erosion mitigation under harsh semi-arid conditions, but with weaker effects than more persistent grass cover on both soil protection and soil quality (Sastre et al., 2017, 2018). However, their efficacy in controlling erosion is not always consistent. This variability is also reflected in the systematic review by Abad et al. (2021), who reported that vineyard cover crops generally improve SOC, infiltration, aggregate stability, and erosion control. However, the same review observed that SOC gains and erosion reduction are usually more consistent under grass-dominated covers, whereas responses under legume covers are more variable. This variability can be partly explained by differences in biomass production, residue persistence, and residue quality. Indeed, legume residues tend to decompose rapidly, because of the low C/N ratio, resulting in less persistent soil cover compared with grasses. For instance, Gaudin et al. (2013) reported that red clover residues, with a C:N ratio between 13.6 and 16.7, released the majority of their nitrogen within 10 weeks, highlighting their fast decomposition rate. Legumes may provide less persistent soil cover, as legume residue are quickly decomposed and legumes produced less residue biomass than other crops. Thus, the reduced soil cover makes soil erosion less consistent than grass or mixed-species systems (Blanco-Canqui et al., 2015). Ramirez-Garcia et al. (2015) compared five cover crop species in Mediterranean semi-arid conditions and found that grasses such as barley, rye, and triticale reached the highest ground cover (83–99%) and biomass production (1226–1928 g m⁻²), with C:N ratios ranging from 27 to 39 and residue quality below 700 mg g⁻¹. These grass traits, together with their slower residue decomposition, are effective for erosion control and catch-crop functions. In contrast, common vetch showed low biomass (425 g m⁻² in the first year and 88 g m⁻² in the second), limited nitrogen uptake (2.4 and 0.7 g N m⁻²), and a low C:N ratio (15–18), but high biological nitrogen fixation (9.8 and 1.6 g N m⁻²) and residue quality around 730 mg g⁻¹. Consequently, common vetch was better suited for soil nitrogen enrichment and green-manure purposes, while grasses, owing to their rapid and persistent cover and high-carbon residues, were more appropriate for mitigating soil erosion (Ramirez-Garcia et al., 2015). Thus, legume-grass mixtures may represent an optimised solution, combining the persistent physical protection of grasses with the fertility-related benefits of legumes. In cherry orchard field with a significant slope in central China, Li et al. (2023) showed that in cover crop plots soil loss was reduced by 30–53% and total P losses by 30–38% compared with no-cover plots. The mixed legume-grass treatment showed the largest overall performance in reducing water, soil, and nutrient losses, suggesting a complementary role of mixed cover systems in erosion and nutrient-loss mitigation (Li et al., 2023). Similarly, Silvestroni et al. (2024) in a Mediterranean hillside vineyard demonstrated that a stable soil cover of approximately 70% can be reached with perennial legume-grass mixture including white clover, bird's-foot trefoil, perennial ryegrass, and red fescue. The legume-enriched mixture increased vine vigor indicators in comparison with natural vegetation: total leaf area rose from 3.7 to 5.1 m² vine⁻¹, pruning weight from 0.57 to 1.02 kg vine⁻¹, and yield increased by 66% in 2021 and 93% in 2022. Although runoff and soil loss were not directly measured, these findings support the multifunctional value of legume-enriched mixtures in perennial systems.

Overall, leguminous cover crops can contribute to erosion reduction. However, their effect is often less reliable and more variable than grass covers when rapid, persistent, and high-biomass soil protection is required. Their strongest role may therefore emerge either in well-established full-surface covers or in mixtures with grasses.

5.1.2. Green manure

Green manure use is recognized as a sustainable agricultural practice which enhances soil structure, increases organic matter content, thereby contributing to long-term soil productivity and environmental sustainability (Iderawumi and Kamal, 2022). Scavo et al. (2022) report that leguminous cover crops such as faba bean, common vetch, lupin and

clovers enhance soil fertility by increasing soil organic matter, nitrogen availability, microbial activity and aggregate stability, making green manuring an effective strategy to improve soil structure and restore fertility under low input conditions. Legume green manures increased soil organic matter by 7.38% in the 0–20 cm layer compared with bare fallow controls. It's also interesting that generic green manure (including legume green manure) reduced bulk density by 5.6% compared with bare fallow. These improvements strengthen soil structure, porosity, and biological activity, thereby increasing the soil's resilience to physical degradation (Ma et al., 2021).

In Oklahoma, a dryland cropping systems, showed that leguminous green manures may offer advantages. Baath et al. (2023) proposed replacing conventional summer fallow periods in the U.S. Southern Great Plains with legumes such as soybean, tepary bean (*Phaseolus acutifolius* A. Gray), and moth bean (*Vigna aconitifolia* (Jacq.) Marechal). These legumes quickly established canopy cover, produced significant biomass (up to 8.9 Mg ha⁻¹). These traits suggest potential to reduce soil exposure during wheat-fallow rotations, although erosion or runoff was not directly measured in the study.

5.1.3. Temporary legume-based intercropping

Intercropping refers to the simultaneous cultivation of two or more crop species within the same field and is used to exploit complementary interactions that may enhance productivity, resource-use efficiency, and environmental performance (Kumawat et al., 2022; Reichmann et al., 2025). In the context of erosion mitigation, legume-based intercropping is mainly relevant when associated species provide rapid soil cover, complementary root systems, and sufficient residue inputs. Gaudin et al. (2013) reported that inter-seeded red clover can improve soil structure and wet aggregate stability, and that rotations including red clover are associated with lower particle detachment and reduced erosion potential. The same review also noted that cover crops and plant residues can improve water penetration and reduce runoff losses, although erosion outcomes were not directly quantified for red clover in that synthesis. Kumawat et al. (2022), summarized evidence showing that legume-based intercropping mitigates soil erosion and runoff although the magnitude of these effects varied among crop combinations and environmental contexts. Reported reductions in soil loss ranged between 24% and 70%, while runoff decreased by 17% to 43%, depending on crop combinations, slope conditions and reference systems. These improvements were attributed to the denser canopy cover, enhanced root biomass, and greater organic matter input of legume-cereal and legume-tuber systems, which can improve soil aggregation, porosity, hydraulic conductivity, and moisture retention. In Jilin province in China, mechanistic evidence was provided by Duan et al. (2026), who showed in a three-year maize-based intercropping study on mollisols that aggregate stability in soybean and peanut (*Arachis hypogaea* L.) strips was higher than in the corresponding legume monocultures. At strip scale intercropping increase the proportion of macroaggregates by 6.4–13.3% and mean weight diameter by 13.9–31.9%, mainly through maize root residues from the previous season and maize root extension during the current season. Thus, legume strips can benefit structurally from belowground interactions with maize, although erosion was not directly measured. Recent field evidence from the Chinese Mollisol region further showed that the effectiveness of legume-based intercropping depends strongly on species selection. In a two-year maize/cover crop intercropping experiment on 4.2° sloping farmland, Chinese milk vetch (*Astragalus sinicus* L.) increased maize grain yield by 12.6% relative to maize sole cropping, while reducing annual runoff by 61.2–70.0% and annual soil loss by 80.6–84.0%, indicating that suitably selected legume living mulches can mitigate erosion without compromising crop productivity (Liu et al., 2025). Conversely, Reichmann et al. (2025), examining legume-based intercropping systems like faba bean/durum wheat (*Vicia faba* L./*Triticum durum* Desf.) and rapeseed-field pea (*Brassica napus* L./*Lathyrus oleraceus* Lam.) intercropping under semi-arid Mediterranean conditions, found no significant improvement

in water-stable macroaggregates, soil organic carbon, or total nitrogen compared to sole cropping, suggesting limitations in short-term effectiveness under intensive tillage and low rainfall conditions (266–380 mm/year).

Overall, these studies indicate that temporary legume-based intercropping can reduce runoff and soil loss when it provides rapid soil cover, favourable canopy architecture, and persistent root activity. However, its effectiveness is strongly context-dependent and may be limited in short-term, intensively managed, or water-limited systems where poor plant establishment can reduce soil cover, residue production, root development, and organic matter inputs.

5.2. Crop management strategies and their effectiveness in erosion control

Legume-based intercropping can reduce runoff and soil loss when it increases ground cover, root inputs, organic matter addition, and aggregate stability, although the magnitude of these effects varies among crop combinations and environmental contexts (Kumawat et al., 2022). These quantitative findings are summarized in Table 3. Singh Jatav and Rajput (2022) indicates that legumes can be integrated through sequential cropping, intercropping, rotations, green manuring, cover cropping, and bio-mulching, with potential benefits for SOC, soil structure, infiltration, and erosion risk reduction. The importance of crop diversification is further supported by Haynes and Beare (1997), who conducted a 12-week greenhouse experiment on a heavily cropped arable soil and found that crop species differed markedly in their effects on aggregate stability, with white clover and lupin showing unexpectedly high aggregate stability relative to their root mass and root length. In particular, lupin increased the mean weight diameter from 0.30 to 0.49 mm and produced a four-times greater increase in fungal hyphal length compared with wheat, suggesting a rapid rhizosphere-mediated effect of lupin on aggregate stabilization and microbial activity. Bhat et al. (2019) described that crop rotations including legumes such as peanut, soybean, mung bean, and chickpea enhance soil fertility, organic matter, and infiltration. Thus, legume-based crop rotations indirectly contribute to halt soil erosion with sufficient soil cover, residue inputs, and improved soil structure. Further, increasing straw mulch application rates from 0 to 6 t ha⁻¹ markedly decreased runoff (from 50% to 1.2%) and soil loss (from 4.83 to 0.05 t ha⁻¹) (Bhat et al. (2019)). Cover crops, including legumes, can contribute to erosion control by maintaining soil cover and improving soil physical and hydrological properties. In perennial systems such as vineyards and orchards, cover crops have been reported to reduce erosion and runoff. Quintarelli et al. (2022) summarized evidence from vineyards showing soil loss reductions from 3.8 to 0.7 Mg ha⁻¹ and annual runoff reductions of 27%. These values refer to cover-crop-based management in general. Further evidence from a three-year field experiment on the slope in the Gansu province, in Chinese Loess Plateau, showed that the erosion-control performance of alfalfa strongly depended on tillage management. Indeed, open-ridging (ridge-furrow planting with open runoff-collecting furrows) reduced runoff by 17.9–37.7% and soil loss by 46.4–75.5%, whereas tied-ridging (furrows interrupted by cross-ridges forming micro-basins) achieved even greater reductions of 22.3–55.5% and 62.1–87.6%, respectively in comparison with conventional tillage management. When combined with biochar amendment, these systems also reduced soil nutrient losses and improved soil water storage, indicating that management configuration can substantially amplify erosion mitigation in alfalfa-based systems (Zhao et al., 2024).

Overall, these studies show that the erosion-mitigating effectiveness of legume-based management strategies depends on species choice, system design, structural soil management, and management intensity.

6. Discussion

This review showed recurring patterns, rather than uniform effects, in the influence of legume-based systems on runoff, soil loss, and

infiltration across Temperate and Mediterranean agroecosystems (Table 2). Runoff reductions ranged from 17 to 43% and soil-loss reductions from 24 to 70% when legume-based covers or intercrops were compared with bare soil, sole cropping, or less protective management systems (Table 3). Soil-loss decreases of up to 13.2 t ha⁻¹ yr⁻¹ have been reported in Mediterranean perennial systems. Although these effects cannot be attributed exclusively to legumes when the treatment combined legume cover with broader changes in vegetation and soil management. Reported C-factor values further illustrate this variability: Pan-European estimates assign dried pulses and protein crops a generic C crop value of 0.32. In vineyards, cover crops consisting exclusively of legumes have shown higher C-factor values, ranging from 0.21 to 0.23, whilst dense mixtures of legumes and grass in vineyards have been reported to reach values of up to 0.015. Among the studies reviewed that compared cereals and legume under semi-arid Mediterranean conditions, barley achieved higher soil surface cover than common vetch (61.7% vs 44.4%), and saturated hydraulic conductivity, Ks, under barley was higher than common vetch (36.1 vs 23.2 mm h⁻¹). Infiltration responses were often positive relative to bare soil or fallow, but the magnitude and direction of legume-grass contrasts remained strongly context-specific. Structural indicators likewise showed mixed rankings depending on site and metric, with some studies reporting slightly lower aggregation under legumes than grass references, while others documented substantial gains relative to no-cover controls.

Quantitative patterns reviewed suggest that the erosion-mitigating role of legumes, emerges from multiple interacting mechanisms, whose expression depends on species traits, management intensity, and environmental context. A complete compilation of the quantitative data contained within this review and relating to legume-based cropping systems can be found in the supplementary materials, specifically in Table S1. The gathered evidence shows that legumes can contribute to erosion control when timely ground protection is achieved and when belowground effects are associated with improvements in infiltration, aggregation, or soil cohesion. However, grasses may match or outperform legumes for short-term surface protection where persistent canopy cover and residue durability dominate erosion processes. Taken together, Tables 2 and 3 suggest a functional interpretation in which legumes and grasses have partially complementary erosion control mechanism. Grasses are often more effective in providing rapid and persistent surface cover, whereas legumes can support soil structure, infiltration, and biological activity. Combining these functions, for example through legume-grass mixtures or by matching cover development with periods of high rainfall erosivity, may improve erosion control under suitable management conditions.

6.1. Comparing legumes and grasses

Legumes and grasses differ in the way they influence erosion-related processes, and their functions are often complementary rather than interchangeable. Grasses often provide short-term erosion control through rapid canopy closure, dense fibrous roots, high biomass production, and more persistent residues, which help reduce raindrop impact, runoff energy, and sediment detachment after termination or senescence (Blanco-Canqui and Ruis, 2020; Capri et al., 2023; Ramirez-Garcia et al., 2015; Vallebona et al., 2016). On the other hand, legumes may also contribute to increase the resistance to erosion more indirectly by modifying rhizosphere conditions, particularly through microbial activity, organic inputs, and improved aggregate stability. However, these pathways are not always directly reflected in measured runoff or soil-loss reductions (Haynes and Beare, 1997; Jones et al., 2009). These changes support soil structural stability. However, their effect is mediated by soil characteristics and crop management. The faster decomposition of legume residues can support nutrient mineralization and microbial activity; despite reducing the persistence of physical soil cover in comparison to grass (Scavo et al., 2022). Therefore, the most suitable species should be selected according to the main

erosion-control objective, whether rapid surface protection, residue persistence, fertility restoration, or longer-term improvement of soil structure.

6.1.1. Comparison of annual legumes and cereals

Annual grasses and legumes differ in several traits influencing erosion-related processes, including canopy development, biomass production, residue persistence, and root architecture. Annual cereal such as barley, rye, oat, triticale and winter wheat develop fibrous root systems, rapid canopy closure, and high early biomass which can favour early soil protection. These traits are associated with improved soil cover and anchorage, which are important for reducing detachment and runoff-driven sediment transport. On the other hand, several annual legumes common vetch, hairy vetch, and faba bean, may provide less rapid or less persistent early soil cover than cereal covers under some conditions, partly because of differences in canopy development, residue quality, and root architecture (Neugschwandtner et al., 2019; Ramirez-Garcia et al., 2015; Wong et al., 2024). In Mediterranean vineyards, durum wheat and grass-containing covers provided stronger soil protection than faba bean-based covers, as indicated by lower C-factor values, whereas faba bean alone showed one of the highest C-factor values and was among the least effective cover crop treatments (Novara et al., 2011). Wagg et al. (2021) further showed that cover-crop functional groups have different effects on soil erosion control depending on the season. In Canada under temperate climate, Wagg et al. (2021)'s full-season field experiment, legumes provided the lowest average growing-season cover, whereas winter cover was greatest under legumes, particularly hairy vetch and red clover. However, the highest overall erosion-control potential was observed under oilseed radish (*Raphanus sativus* var. *Oleiformis* L.), tillage radish (*Raphanus sativus* L.), and Italian ryegrass. Other evidence indicate that annual grasses often provide a better physical erosion control because their vigorous growth, rapid canopy development, and slower-decomposing residues ensure more persistent surface protection, whereas annual legumes contribute more strongly to nitrogen inputs and biological soil fertility (Koudahe et al., 2022; Van Eerd et al., 2023).

Annual legumes can support soil fertility and soil biological activity, however their short-term erosion-control role may be weaker and slower than grasses/cereal covers, as legumes produce less biomass at the early stages or have a slower canopy closure. This pattern is supported by long-term measurements on 16% slopes in north-eastern Romania, where annual legume crops reduced runoff and soil loss compared with row crops, but remained less protective than wheat or perennial grasses. Runoff was 15.4 mm in pea and 15.9 mm in soybean, compared with 29.7–31.2 mm in maize and sunflower, while annual soil loss was 1.84–1.93 t ha⁻¹ in pea and soybean, 4.33 t ha⁻¹ in bean, 0.56 t ha⁻¹ in wheat, and only 0.21 t ha⁻¹ in second-year perennial (Bucur et al., 2011). Thus, belowground improvements in infiltration or aggregation may be insufficient to reduce runoff and soil loss when aboveground soil protection is still limited. In particular, if erosive rainfall occurs before annual legumes have developed enough canopy cover or residue protection, the soil surface may remain exposed and, despite improvements in soil physical properties, be susceptible to erosion. Therefore, annual grass or cereal cover are more suited for immediate erosion control where intense rainfall, slope, or short bare-soil windows where rapid and persistent cover is essential. Annual legumes, in contrast, may be most useful when they are integrated into systems designed to combine seasonal soil protection with fertility restoration and longer-term improvement of soil structure.

6.1.2. Perennial legumes and perennial grasses

Perennial legumes and grasses influence erosion-related processes through complementary mechanisms, particularly in relation to surface cover, root reinforcement, infiltration, and soil structural development. Fibrous-rooted perennial grasses can provide strong near-surface reinforcement by increasing root density in the upper soil layer, physically

binding soil particles, and promoting aggregate stability. In a comparative screening of cover crop species for orchard and vineyard floor management, several grasses, including sheep fescue and red fescue, showed high root length density and root dry weight, traits associated with large macroaggregate formation and higher mean weight diameter (Capri et al., 2023). Perennial legumes also contribute to erosion mitigation, with different mechanism among species. Sainfoin reduced runoff and soil loss in semi-arid field plots, with protection increasing as plant cover developed over successive years (Yüksek and Yüksek, 2015), whereas alfalfa can enhance infiltration through deep tap-root systems and decayed root channels (Guo et al., 2019). The belowground effects of perennial legumes should be interpreted as process-specific rather than uniformly positive. In sainfoin plots, runoff and soil loss declined mainly as vegetation cover increased over time (Yüksek and Yüksek, 2015), whereas alfalfa root channels can enhance preferential flow and infiltration after root decay (Guo et al., 2019). Rainfall-simulation evidence from the Loess Plateau in Shaanxi province, further showed that grassland effects on runoff, soil loss, and infiltration depend on species and planting age: gramineous grasslands with dense fibrous roots achieved stronger sediment reduction and runoff maintenance than alfalfa and white clover, although white clover performed better than alfalfa among the legumes (Liu et al., 2019). At the system scale, perennial forage legumes may also support soil fertility and structural recovery through nitrogen inputs, residue return, and organic matter accumulation, although much of this evidence is synthesis-based rather than derived from direct erosion measurements (Tomić et al., 2023).

This perennial legume effect is strongly context-dependent. In the Chinese Loess Plateau, perennial ryegrass provided stronger shallow reinforcement under drier and steeper conditions, whereas alfalfa became more effective under wetter conditions and gentler slopes (Cao et al., 2025). This broader pattern is consistent with the trait-based analysis of Wang et al. (2026), who showed that fibrous-rooted herbaceous species had a higher comprehensive root-system parameter and a lower relative soil detachment capacity than tap-rooted species in the Loess Plateau. These differences, however, cannot be interpreted in terms of root architecture alone, because soil detachment was controlled by the combined effects of root surface area density, root mass density, root cohesion, and root topology, and was also influenced by soil disturbance and management. Jia et al. (2024) showed that root effects on infiltration reflect the balance between partly opposing processes, including the formation of non-capillary pores, changes in bulk density, organic matter accumulation, root exudation, and root decay; notably, enhanced non-capillary porosity accounted for more than 50% of the positive effect on infiltration, whereas root-induced increases in soil organic matter could partly offset this benefit. Perennial grasses may provide more immediate erosion control where shallow reinforcement and persistent surface protection are the main requirements. Instead, perennial legumes, may contribute more progressively through cover development, root-channel formation, organic inputs, and improvements in fertility and soil structure. Their relative effectiveness therefore depends less on species identity alone than on the interaction between root architecture, vegetation age, cover development, soil moisture, slope conditions, and the management objective, whether short-term soil protection or longer-term soil restoration.

6.1.3. Effectiveness of legume-grain mixes vs. pure legume stands

Mixed sowings of legumes and grasses can provide stronger and more multifunctional soil protection than pure legume stands in some systems, particularly when grass components increase persistent cover and residue protection while legumes contribute to nutrient cycling. Field experiments and reviews supports a functional-complementary interpretation: grass components can strengthen physical soil protection through cover, residues, and fibrous roots, whereas legumes can contribute to nitrogen inputs and biological fertility. Where these functions are well balanced, mixtures may enhance erosion control and soil resilience relative to pure legume covers (Blanco-Canqui and Ruis,

2020; Li et al., 2023; Novara et al., 2011). Blanco-Canqui et al. (2015) argued that mixed-species cover crops may be more multifunctional than single-species stands because different species provide complementary services, with grasses contributing rapid cover and soil protection and legumes contributing biological nitrogen inputs. However, they also noted that the evidence base for erosion responses across mixture types remains limited. Scavo et al. (2022) further support the broader rationale for mixed cover crops, noting that mixtures can combine complementary functions, including nitrogen supply from legumes and carbon inputs from grasses, and may improve nutrient retention when species composition and management are well matched. Mechanistic evidence from maize-legume intercropping further supports the role of interspecific root interactions in soil aggregation. Duan et al. (2026) showed that maize-legume intercropping increased aggregate stability in legume strips through both current-season maize root extension and residual root effects from the previous season, although this evidence refers to maize-legume systems rather than grass-legume cover-crop mixtures. This interpretation is also consistent with forage-system evidence showing that legume-grass mixtures can improve nitrogen transfer, biomass production, and stand persistence, although these outcomes provide indirect rather than direct evidence for erosion mitigation (Tomić et al., 2023).

From a process-based perspective, the advantage of effective legume-grass mixtures is better interpreted as functional complementarity rather than as a simple additive effect. Grasses commonly strengthen surface protection and near-surface reinforcement through rapid cover, persistent residues, and fibrous roots, whereas legumes often contribute through nitrogen inputs, residue quality, biological activity, and, in tap-rooted species, deeper soil structuring. The combination of these groups (i.e. grasses and legumes) can increase the temporal continuity of soil cover and broaden the depth range over which roots contribute to soil stabilization. In an engineered steep-slope restoration context, Meng et al. (2025) showed that grass-legume mix grassland increased soil shear strength and anti-erodibility relative to untreated natural slopes, with anti-erodibility increasing by 31.29% on sunny slopes and 47.44% on shady slopes. However, this response reflected mechanical stabilization and reduced hydraulic conductivity rather than a simple improvement in infiltration or aggregate stability. These mechanisms suggest that mixed stands may provide more robust soil protection than pure legume covers under some conditions, however direct effects on runoff and detachability remain dependent on rainfall regime, growth stage, species composition, and management. Nevertheless, these benefits remain strongly dependent on species composition, relative dominance within the mixture, establishment timing, and management practices that influence biomass distribution, residue persistence, and seasonal cover continuity.

6.2. Crop management strategies and their effectiveness in erosion control

The capacity of leguminous species to mitigate soil erosion is strongly conditioned by crop management strategies such as sowing time, tillage system, biomass and residue handling, and integration into rotations or intercropping systems. Seeding time plays a key role because it determines whether legume cover crops establish sufficient soil cover before periods of elevated rainfall erosivity. This flexibility may allow cover establishment to be aligned with periods of elevated rainfall erosivity. In Mediterranean-type environments, including Italian and central-Italian conditions, erosivity is often concentrated in summer and autumn events characterized by high rainfall intensity, making the timing of cover establishment critical for erosion control (Borrelli et al., 2016; Todisco et al., 2025). Furthermore, the extent, spatial configuration, and timing of plant cover, together with agronomic soil management, determine whether legume-based systems result in a measurable reduction in erosion. This configuration effect was clearly illustrated by Zuazo et al. (2020) in Mediterranean rainfed olive orchards on 20% slopes. Among the individual treatments, the no-tillage system with

legume cover over the entire soil surface (LEC) achieved the lowest mean soil loss ($0.32 \text{ t ha}^{-1} \text{ year}^{-1}$), outperforming both conventional tillage without plant cover ($3.25 \text{ t ha}^{-1} \text{ year}^{-1}$) and legume-strip configurations under minimum tillage and no-tillage (2.15 and $2.50 \text{ t ha}^{-1} \text{ year}^{-1}$, respectively). By contrast, no-tillage with spontaneous vegetation strips also performed effectively ($0.54 \text{ t ha}^{-1} \text{ year}^{-1}$), indicating that the anti-erosive value of legumes was greatest when they provided continuous soil cover rather than partial strip protection.

The adoption of reduced or no-tillage systems can increase the erosion-reducing potential of legume-based covers when they preserve surface residues, maintain soil structure, and support continuous cover. For instance, faba bean grown under no-till conditions on Vertisols developed a more extensive and deeper root system compared with conventional tillage, suggesting a mechanism through which reduced disturbance of soil may improve soil exploration and contribute to structural functioning (Muñoz-Romero et al., 2011). Bowles et al. (2017) showed that legume-based cover crops under conservation or no-tillage increased AMF colonization. This supports a biological mechanism potentially relevant to soil structure, but it does not provide direct evidence of erosion reduction. In a two-year field experiment on sloping land in central Greece, an autumn triticale-field pea intercrop followed by spring sunflower (*Helianthus annuus* L.) reduced surface runoff by 85% when combined with no-tillage and contour planting, compared with a no-plant no-tillage control (Molla et al., 2022). This supports the effectiveness of combining crop cover, rotation, contour planting, and reduced disturbance, although the specific contribution of the legume component was not isolated. Evidence from Mediterranean perennial systems also suggest that legume integration can support soil-conserving agronomic management. Mantino et al. (2016) reported that alfalfa overseeding into mature switchgrass stands under rainfed, low-input conditions increased total annual biomass by about 19% compared with pure switchgrass. Although runoff and soil loss were not measured, the higher biomass and perennial cover suggest a management configuration that may reduce soil exposure.

The timing of cover crop termination and subsequent residue management are critical issues which has long been recognized in winter cover crop systems. Dabney et al. (2001) emphasized that the effects of cover crops on soil and water conservation depend strongly on species selection, planting date, and killing date, since cereal residues with high C:N ratios tend to immobilize N whereas legume residues decompose more rapidly; under low residual soil N, grass-legume mixtures may provide a more balanced system by combining N scavenging with biologically fixed N and greater carbon inputs. Garcia et al. (2018) emphasized that the performance of vineyard service crops should be interpreted as the outcome of a management-driven balance between ecosystem services and disservices, shaped by species choice, spatial distribution, sowing date, maintenance, and termination time in relation to local pedoclimatic conditions and production objectives. A broader systems analysis by Thomas et al. (2018) similarly showed that management practices directly targeting ground cover had the strongest influence on erosion risk, and that the timing of interventions during high-erosivity periods was critical. Pala et al. (2000) highlighted that legumes in rainfed cereal rotations help protect soil from erosion, add organic matter, fix nitrogen, and reduce the build-up of cereal diseases and weeds, thereby improving the performance of subsequent cereal crops. Decision-support approaches can help translate these principles into management planning: crop-rotation optimization models can incorporate legumes such as alfalfa and clover as soil-improver crops while balancing sustainability and farm profitability, whereas meta-analytical tools such as the Soil Health Calculator can estimate expected cover-crop effects on soil health indicators. However, these tools should be interpreted as decision-support aids rather than direct predictors of erosion control (Benini et al., 2025; Jian et al., 2020a, 2020b).

Although legumes can contribute to erosion control through root development, organic inputs, nutrient cycling, and effects on soil

structure, these functions translate into measurable erosion reduction only when legumes are integrated into well-timed and well-managed cropping systems. Using legumes within diverse cropping systems help for building ecosystem resilience. [Kaye and Quemada \(2017\)](#) indicate that cover crops can increase adaptive capacity under climate change by reducing vulnerability to erosion from extreme rainfall, improving water management options, and retaining nitrogen; within this framework, legumes contribute mainly through biological nitrogen inputs and mixture functionality rather than through erosion control alone.

Across the available evidence, crop management appears to be at least as important as legume species identity in determining erosion outcomes. The capacity of legumes to reduce runoff and soil detachability depends largely on whether sufficient soil cover is present during periods of peak rainfall erosivity, a condition strongly controlled by sowing date, tillage regime, and residue persistence. Therefore, legumes should not be considered standalone erosion-control solutions, but rather functional components whose contribution becomes evident when they are embedded within coherent, system-level management strategies. This interpretation is consistent with the broader synthesis of [Franzluebbers et al. \(2014\)](#), who concluded that reintegrating perennial grasses and legumes into agricultural systems can improve soil organic matter, infiltration, water quality, and erosion control, particularly within pasture-crop rotations. Similarly, [Haruna et al. \(2025\)](#) showed that three years of no-till multispecies cover cropping increased soil organic carbon by 54%, volumetric water content by 17%, and microbial biomass by 22% relative to no-cover management, supporting the role of diversified cover-crop systems in improving soil health and resilience indicators, even though runoff and soil loss were not measured and the specific contribution of legumes could not be isolated.

6.3. Model parameterization limits for legume-based erosion control: WEPP, EUROSEM, and RUSLE

Model interpretation is important because legume-based systems are not represented only by species identity, but by time-varying traits such as canopy development, root growth, residue persistence, soil cover, and management operations. These differences matter because WEPP, EUROSEM, and RUSLE represent vegetation effects at different temporal scales, from continuous simulation to event-scale parameterization and long-term average soil-loss estimation.

WEPP is a widely used physically based continuous simulation model which estimates soil erosion and sediment yield through the simulation of as infiltration, runoff, tillage effects, residue decomposition, root biomass and sediment transport ([Flanagan and Nearing, 1995](#)). While WEPP represents a step forward over empirical approaches by simulating detachment, transport, and deposition processes mechanistically, its application to legume-based systems remains constrained. The plant-growth component simulates biomass accumulation, canopy development, root growth, leaf area index, and senescence. In the table of parameters relating to the growth of cultivated plants provided in the WEPP documentation, there are only a limited number of leguminous crops with explicit parameters, specifically soya, alfalfa and peanut ([Flanagan and Nearing, 1995](#)). Common cover legumes such as vetch, faba bean, and clover are not included as default cropland plant-growth parameter entries in the available WEPP documentation. This forces users to develop or adapt parameters based on similar crops or external datasets. Uncertainty increases if WEPP is applied to legume-based rotations without local calibration. In addition, the residue decomposition sub-model requires crop-specific inputs for variables like decomposition rate (ORATE), residue fragility, and cutting height (H-cut). In the WEPP residue-decomposition sub-model, soybean and peanut residues are classified as fragile, whereas alfalfa is classified as non-fragile. The default decomposition rates assigned to legume residues are generally higher than those assigned to cereal residues, making simulated soil protection sensitive to how residue persistence and surface cover are parameterized. The WEPP soil component includes inter-rill erodibility,

rill erodibility, critical shear stress, and effective hydraulic conductivity, all of which can be affected by soil disturbance, live and dead root biomass, incorporated residues, sealing and crusting, compaction, and consolidation. Consequently, although WEPP enables process-based simulation under varying management practices, its accuracy in legume-based systems is hindered by the absence of adequate crop-specific input data ([Flanagan and Nearing, 1995](#)).

In contrast, EUROSEM is an event-based, physically based model designed to simulate runoff and sediment transport during individual storm events. It models rainfall interception, infiltration, surface detention, overland flow, and sediment detachment by both raindrop impact and runoff shear ([Morgan et al., 1998](#)). Vegetation is represented through parameter and inputs such as vegetation cover, interception storage, canopy height, stem angle, leaf shape, basal stem area, and related surface-cover parameters. However, unlike WEPP, EUROSEM does not simulate vegetation development dynamically; as users must input all vegetation data for each rainfall event. This can make simulations sensitive to the phenological stage chosen for parameterization, especially for legumes whose canopy, root traits, and soil-surface effects change during growth. In a rainfall-simulation field experiment with soybean on a gently sloping farmland, [Yao et al. \(2025\)](#) showed that runoff and sediment rates decreased from seeding to final flowering or grain formation and increased again toward ripening, following changes in canopy cover, root traits, and near-surface soil properties. A small-plot laboratory study with red clover reached a similar conclusion: vegetated plots reduced runoff and soil loss relative to bare controls, during the flowering stage the crop provided the largest erosion control, mainly by limiting soil detachment rather than reducing runoff ([Naderi Marangalo et al., 2025](#)). Although the EUROSEM manual provides vegetation parameter values for several crop and pasture categories, these values are static and generalized rather than stage-specific ([Morgan et al., 1998](#)).

RUSLE remains one of the most widely used tools for estimating long-term average annual soil loss. It is based on the original USLE developed by [Wischmeier and Smith \(1978\)](#) and includes revised factors and updated databases. The model calculates erosion as the product of rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover-management (C), and support practices (P). Among these, the C-factor is crucial in quantifying the protective effect of vegetation and management. In RUSLE, the C-factor reflects prior land use, surface cover, crop canopy, surface roughness, and, in some applications, soil moisture, while also accounting for temporal changes in crop cover, root mass, tillage, and residue decomposition ([Renard et al., 1997](#)). This structure allows RUSLE to incorporate several vegetation and management effects. However, applications relying on a single tabular or annual average C-factor for a legume crop may oversimplify its seasonal erosion-control behaviour, particularly when erosion risk depends on the alignment between crop development and erosive rainfall. The phenological responses reported for soybean and red clover reinforce this point, because protection varied substantially across growth stages rather than remaining constant throughout the crop cycle ([Naderi Marangalo et al., 2025](#); [Yao et al., 2025](#)). Historical USLE/RUSLE crop-management databases provide values for several major crops, but many legume cover crops and mixtures still lack locally calibrated C-factor values, requiring users to rely on proxy values or local calibration ([Renard et al., 1997](#); [Wischmeier and Smith, 1978](#)). However, many cover legumes are not included in standard C-factor datasets, requiring the user to interpolate or estimate values from analogous species. As a result, simplified RUSLE applications in legume-based conservation systems may be constrained by limited crop-specific calibration and by the use of static or annual average C-factor values.

These modelling constraints help explain why discrepancies among studies on legumes, runoff, and soil loss may reflect model structure and parameterization as well as field conditions. Biological traits and management practices remain central, but their effects can be simplified or distorted when models use incomplete legume representation,

generalized residue dynamics, or static vegetation parameters. Future applications should therefore prioritize stage-specific parameterization of legume cover, root development, residue persistence, and management operations, especially in systems where erosive rainfall occurs before, during, or after peak legume protection.

6.4. Recommendations and future directions

6.4.1. Practical recommendations for erosion control with legumes

Two main practical recommendations emerge from this review.

1. Design legume-based covers according to local erosion-risk periods and management windows.
 - Legume-based systems should be planned to have effective soil cover prior rainfall erosivity peaks. Site-specific erosivity patterns can help identify high-risk periods, particularly in Mediterranean environments where rainfall erosivity is strongly seasonal (Borrelli et al., 2016; Todisco et al., 2025)
 - Species selection should also consider crop-cycle compatibility, termination requirements, and residue persistence within the rotation (Murphy-Bokern et al., 2017). Annual mixes, such as common vetch (*Vicia sativa* L.) with oat (*Avena sativa* L.) or hairy vetch (*Vicia villosa* Roth) with rye (*Secale cereale* L.), may provide seasonal soil protection when rapid cover is required.
 - Perennial mixtures, such as red clover (*Trifolium pratense* L.) with tall fescue (*Lolium arundinaceum* (Schreb.) Darbysh.) or alfalfa (*Medicago sativa* L.) with perennial ryegrass (*Lolium perenne* L.), may be more suitable for orchards, vineyards, and marginal lands where persistent cover and reduced soil disturbance are priorities.
 - In all cases, the combination of the species should be related with their growth cycles, termination timing, and soil-preparation requirements.
2. Combine legumes with soil conservation management practices
 - Legume-based covers are more likely to support erosion control when they are integrated with improved residue management
 - No-till and reduced tillage can preserve surface residues, reduce bare-soil exposure, and support belowground biological functioning. Their effects on soil erosion depend on soil condition and management context (Bowles et al., 2017; Hu et al., 2023; Muñoz-Romero et al., 2011)

Management should also support arbuscular mycorrhizal fungi and rhizobial symbioses, especially in low-input systems. Reduced disturbance and continuous cover can help preserve mycorrhizal networks, while rhizobial inoculation may support nodulation, biological nitrogen fixation, root development, and soil fertility in legume-based systems (Bowles et al., 2017; Kabalan et al., 2022; Sun and Shahrajabian, 2022).

These biological mechanisms can support aggregation and soil resilience, but their contribution to erosion control should be interpreted as indirect unless runoff or soil-loss outcomes are measured.

6.4.2. Research priorities for legume-based erosion control

Despite the growing body of evidence on the role of legumes in erosion control, major gaps remain in quantifying their effects and translating empirical results into reliable model parameters. In the erosion models considered here, legume-specific traits are often represented through a limited number of crop entries, generalized vegetation parameters, or proxy values. As a result, simulations may fail to capture the full contribution of time-dependent biological processes, including residue decomposition, root reinforcement, and symbiotic interactions. Addressing these limitations requires an integrated research effort combining long-term field data, trait-based measurements, and dynamic modelling approaches:

1. Stronger field evidence for legume-specific erosion effects should be sought

These data would help derive species- and management-specific values for C-factor dynamics, canopy development, root biomass, root length density, root diameter distribution, macro-porosity-related indicators, residue persistence, and stage-specific vegetation inputs.

2. Improved representation of dynamic legume processes in erosion models.

A second knowledge gap concerns the time-dependent biological processes that are still poorly represented in erosion modelling. Legume-specific data are needed on residue decomposition kinetics, C ratio dynamics, lignin content, root longevity, dead-root turnover, post-harvest pore formation, and changes in surface cover after mowing, frost, incorporation, senescence, or termination. These measurements would help clarify how pure legumes, grasses, and legume-grass mixtures differ in their ability to maintain soil protection over time and influence longer-term soil structure. AMF-related indicators, including glomalin-related soil proteins, should also be tested against aggregate stability and measured erosion outcomes, rather than treated only as indirect proxies.

7. Conclusion

This review shows that herbaceous legumes can contribute to soil erosion mitigation, but their effects are strongly context-dependent. Legumes can reduce erosion-related processes by increasing soil cover, improving infiltration, supporting aggregation, enhancing biological activity, and contributing to root-mediated soil stabilization. Within the RUSLE framework, their effects are most clearly expressed through the cover-management factor, while their influence on soil erodibility should be interpreted more cautiously because most studies measured related soil properties rather than the K factor directly. Legumes are not consistently superior to grasses or cereal-based covers. Grasses often provide more effective protection in the short term, as they offer faster canopy development, greater biomass production, more persistent residues, and stronger short-term protection against raindrop impact, runoff, and sediment detachment. On the other hand, legumes contribute to soil erosion through nitrogen inputs, residue quality, rhizosphere activity, root-channel formation, and longer-term improvements in soil structure. Their erosion-control value therefore depends on local rainfall erosivity, soil conditions, and management objectives. Legumes have the largest soil erosion control effect when they are integrated into well-managed systems, including legume-grass mixes, reduced or no-tillage, improved residue management and with locally adapted sowing or termination dates. Factor affecting legume soil erosion mitigation potential are: poor establishment, rapid residue decomposition, partial cover, water limitation, or mismatch between crop phenology and erosive rainfall peak. Overall, legumes should be considered functional components of diversified erosion-control strategies rather than standalone solutions. Future research should prioritize long-term field trials, direct runoff and soil-loss measurements, and improved model parameterization for legume-based systems.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI's ChatGPT (GPT-4/GPT-5) to support the writing process by assisting with language editing and synthesis of scientific literature. The tool was employed to improve clarity, structure, and consistency in academic writing. All content generated by the AI was carefully reviewed, verified against original sources, and edited as needed by the author. The authors take full responsibility for the accuracy and integrity of the published manuscript.

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CRediT authorship contribution statement

Simone Sorini: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Pietro Goglio:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Validation, Writing – review & editing. **Francesca Todisco:** Conceptualization, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pietro Goglio reports financial support was provided by European Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2026.130320>.

Data availability

Data will be made available on request.

References

- Abad, J., Mendoza, I.H. de, Marín, D., Orcaray, L., Santesteban, L.G., 2021. Cover crops in viticulture. A systematic review: implications on soil characteristics and biodiversity in vineyard. *OENO One* 55, 295–312. <https://doi.org/10.20870/oeno-one.2021.55.1.3599>.
- Acharya, P., Ghimire, R., Idowu, O.J., Shukla, M.K., 2024. Cover cropping enhanced soil aggregation and associated carbon and nitrogen storage in semi-arid silage cropping systems. *Catena* 245, 108264. <https://doi.org/10.1016/j.catena.2024.108264>.
- Ashiaqbor, G., Forkuo, E.K., Laari, P., Aabeyir, R., 2013. Modelling soil erosion using RUSLE and GIS tools. *Int. J. Rem. Sens.* 2, 7–17.
- Baath, G.S., Sarkar, S., Northup, B.K., Sapkota, B.R., Gowda, P.H., Flynn, K.C., 2023. Summer pulses as sources of green manure and soil cover in the U.S. Southern Great Plains. *Crop Environ* 2, 66–74. <https://doi.org/10.1016/j.crope.2023.04.001>.
- Baiamonte, G., Minacapilli, M., Novara, A., Gristina, L., 2019. Time scale effects and interactions of rainfall erosivity and cover management factors on vineyard soil loss erosion in the semi-arid area of Southern sicily. *Water* 11, 978. <https://doi.org/10.3390/w11050978>.
- Bais, H.P., Park, S.-W., Weir, T.L., Callaway, R.M., Vivanco, J.M., 2004. How plants communicate using the underground information superhighway. *Trends Plant Sci.* 9, 26–32. <https://doi.org/10.1016/j.tplants.2003.11.008>.
- Bais, H.P., Weir, T.L., Perry, L.G., Gilroy, S., Vivanco, J.M., 2006. The role of root exudates in rhizosphere interactions with plants and other organisms. *Annu. Rev. Plant Biol.* 57, 233–266. <https://doi.org/10.1146/annurev.arplant.57.032905.105159>.
- Benini, M., Detti, P., Nerozzi, L., 2025. Optimization models and algorithms for sustainable crop planning and rotation: an arc flow formulation and a column generation approach. *Omega* 135, 103320. <https://doi.org/10.1016/j.omega.2025.103320>.
- Bertin, C., Yang, X., Weston, L.A., 2003. The role of root exudates and allelochemicals in the rhizosphere. *Plant Soil* 256, 67–83. <https://doi.org/10.1023/A:1026290508166>.
- Bhat, S.A., Dar, M.U.D., Meena, R.S., 2019. Soil erosion and management strategies. In: Meena, R.S., Kumar, S., Bohra, J.S., Jat, M.L. (Eds.), *Sustainable Management of Soil and Environment*. Springer, Singapore, Singapore, pp. 73–122. https://doi.org/10.1007/978-981-13-8832-3_3.
- Blanco-Canqui, H., Ruis, S.J., 2020. Cover crop impacts on soil physical properties: a review. *Soil Sci. Soc. Am. J.* 84, 1527–1576. <https://doi.org/10.1002/saj2.20129>.
- Blanco-Canqui, H., Shaver, T.M., Lindquist, J.L., Shapiro, C.A., Elmore, R.W., Francis, C. A., Hergert, G.W., 2015. Cover crops and ecosystem services: insights from studies in temperate soils. *Agron. J.* 107, 2449–2474. <https://doi.org/10.2134/agronj15.0086>.
- Borrelli, P., Diodato, N., Panagos, P., 2016. Rainfall erosivity in Italy: a national scale spatio-temporal assessment. *Int. J. Digit. Earth* 9, 835–850. <https://doi.org/10.1080/17538947.2016.1148203>.
- Bowles, T.M., Jackson, L.E., Loeher, M., Cavnano, T.R., 2017. Ecological intensification and arbuscular mycorrhizas: a meta-analysis of tillage and cover crop effects. *J. Appl. Ecol.* 54, 1785–1793. <https://doi.org/10.1111/1365-2664.12815>.
- Brady, N.C., Weil, R.R., 2016. *The Nature and Properties of Soils*, fifteenth ed., global edition. ed, Pearson global edition. Pearson, Harlow, England London New York.
- Bronick, C.J., Lal, R., 2005. Soil structure and management: a review. *Geoderma* 124, 3–22. <https://doi.org/10.1016/j.geoderma.2004.03.005>.
- Bucur, D., Jitareanu, G., Ailincăi, C., 2011. Effects of long-term soil and crop management on the yield and on the fertility of eroded soil. *J. Food Agric. Environ.* 9, 207–209.
- Burton, J.W., 1997. Soyabean (*Glycine max* (L.) Merr.). *Field Crops Res.* 53, 171–186. [https://doi.org/10.1016/S0378-4290\(97\)00030-0](https://doi.org/10.1016/S0378-4290(97)00030-0).
- Cao, Y., Su, X., Zhou, Z., Liu, J., Chen, M., Wang, N., Zhu, B., Wang, P., Liu, F., 2025. Effects of root traits on shear performance of root-soil complex and soil reinforcement in the Loess Plateau. *Soil Tillage Res.* 252, 106625. <https://doi.org/10.1016/j.still.2025.106625>.
- Capri, C., Gatti, M., Fiorini, A., Ardeni, F., Tabaglio, V., Poni, S., 2023. A comparative study of fifteen cover crop species for orchard soil management: water uptake, root density traits and soil aggregate stability. *Sci. Rep.* 13, 721. <https://doi.org/10.1038/s41598-023-27915-7>.
- Comino, E., Marengo, P., Rolli, V., 2010. Root reinforcement effect of different grass species: a comparison between experimental and models results. *Soil Tillage Res.* 110, 60–68. <https://doi.org/10.1016/j.still.2010.06.006>.
- Cui, Z., Huang, Z., Liu, Y., López-Vicente, M., Wu, G.-L., 2022. Natural compensation mechanism of soil water infiltration through decayed roots in semi-arid vegetation species. *Sci. Total Environ.* 819, 151985. <https://doi.org/10.1016/j.scitotenv.2021.151985>.
- Cui, Z., Wu, G.-L., Huang, Z., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. *J. Hydrol.* 578, 124023. <https://doi.org/10.1016/j.jhydrol.2019.124023>.
- Dabney, S.M., Delgado, J.A., Reeves, D.W., 2001. Using winter cover crops to improve soil and water quality. *Commun. Soil Sci. Plant Anal.* 32, 1221–1250. <https://doi.org/10.1081/CSS-100104110>.
- De Baets, S., Torri, D., Poesen, J., Salvador, M.P., Meersmans, J., 2008. Modelling increased soil cohesion due to roots with EUROSEM. *Earth Surf. Process. Landf.* 33, 1948–1963. <https://doi.org/10.1002/esp.1647>.
- Duan, X.-Y., Surigaoe, S., Du, Y.-H., Fu, D.-H., Yang, H., Yang, X., Ma, H.-Y., Zhou, H., Christie, P., Fornara, D., Zhang, W.-P., Li, L., 2026. Interspecific interactions increase soil aggregate stability through altered root traits in long-term legume/maize intercropping. *Soil Tillage Res.* 255, 106808. <https://doi.org/10.1016/j.still.2025.106808>.
- Ellison, W., 1947. Soil erosion studies. Part I. *Agric. Eng.* 28, 145–146.
- Çerçioğlu, M., Udawatta, R.P., Anderson, S.H., 2025. Use of cover crops for sustainable management of soil condition and health: a review. *Soil Secur.* 18, 100177. <https://doi.org/10.1016/j.soisec.2025.100177>.
- Fang, Y., Wang, J., Li, F., Zhang, R., Wang, K., Shanguan, Z., 2024. Contribution of root and soil properties on soil infiltration capacity of gramineous and leguminous forages during root decomposition. *Catena* 246, 108388. <https://doi.org/10.1016/j.catena.2024.108388>.
- Flanagan, D.C., Nearing, M.A., 1995. *USDA-Water Erosion Prediction Project: Hillslope Profile and Watershed Model Documentation* (NSERL Report No. 10). USDA-ARS National Soil Erosion Research Laboratory, West Lafayette.
- Franzluebbers, A.J., Sawchik, J., Taboada, M.A., 2014. Agronomic and environmental impacts of pasture–crop rotations in temperate North and South America. *Agric. Ecosyst. Environ.* 190, 18–26. <https://doi.org/10.1016/j.agee.2013.09.017>.
- Gabriel, J.L., García-González, I., Quemada, M., Martín-Lammerding, D., Alonso-Ayuso, M., Hontoria, C., 2021. Cover crops reduce soil resistance to penetration by preserving soil surface water content. *Geoderma* 386, 114911. <https://doi.org/10.1016/j.geoderma.2020.114911>.
- Gao, G.Y., Fu, B.J., Lü, Y.H., Liu, Y., Wang, S., Zhou, J., 2012. Coupling the modified SCS-CN and RUSLE models to simulate hydrological effects of restoring vegetation in the Loess Plateau of China. *Hydrol. Earth Syst. Sci.* 16, 2347–2364. <https://doi.org/10.5194/hess-16-2347-2012>.
- García-González, I., Hontoria, C., Gabriel, J.L., Alonso-Ayuso, M., Quemada, M., 2018. Cover crops to mitigate soil degradation and enhance soil functionality in irrigated land. *Geoderma* 322, 81–88. <https://doi.org/10.1016/j.geoderma.2018.02.024>.
- García, L., Cellette, F., Gary, C., Ripoché, A., Valdés-Gómez, H., Metay, A., 2018. Management of service crops for the provision of ecosystem services in vineyards: a review. *Agric. Ecosyst. Environ.* 251, 158–170. <https://doi.org/10.1016/j.agee.2017.09.030>.

- Gaudin, A.C.M., Westra, S., Loucks, C.E.S., Janovick, K., Martin, R.C., Deen, W., 2013. Improving resilience of Northern field crop systems using inter-seeded red clover: a review. *Agronomy* 3, 148–180. <https://doi.org/10.3390/agronomy3010148>.
- Gessesse, B., Bewket, W., Bräuning, A., 2015. Model-based characterization and monitoring of runoff and soil erosion in response to land use/Land cover changes in the modjo watershed, Ethiopia. *Land Degrad. Dev.* 26, 711–724. <https://doi.org/10.1002/ldr.2276>.
- Giller, K.E., Andersson, J.A., Corbeels, M., Kirkegaard, J., Mortensen, D., Erenstein, O., Vanlauwe, B., 2015. Beyond conservation agriculture. *Front. Plant Sci.* 6. <https://doi.org/10.3389/fpls.2015.00870>.
- Guo, L., Liu, Y., Wu, G.-L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.-Q., Tian, F.-P., He, H., 2019. Preferential water flow: influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. *J. Hydrol.* 578, 124019. <https://doi.org/10.1016/j.jhydrol.2019.124019>.
- Guo, Z.-B., Yan, G., Zhang, R.-H., Li, F.-M., Zeng, Z.-X., Liu, H., 2010. Improvement of soil physical properties and aggregate-associated C, N, and P after Cropland was converted to grassland in semiarid Loess Plateau. *Soil Sci.* 175, 99. <https://doi.org/10.1097/SS.0b013e3181cda54a>.
- Haichar, F. el Z., Santaella, C., Heulin, T., Achouak, W., 2014. Root exudates mediated interactions belowground. *Soil Biol. Biochem.* 77, 69–80. <https://doi.org/10.1016/j.soilbio.2014.06.017>.
- Hartwig, N.L., Ammon, H.U., 2002. Cover crops and living mulches. *Weed Sci.* 50 (2), 688–699. [https://doi.org/10.1614/0043-1745\(2002\)050%5B0688:AIACCA%5D2.0.CO](https://doi.org/10.1614/0043-1745(2002)050%5B0688:AIACCA%5D2.0.CO).
- Haruna, S.I., Anderson, S.H., Udawatta, R.P., Gantzer, C.J., Phillips, N.C., Cui, S., Gao, Y., 2020. Improving soil physical properties through the use of cover crops: a review. *Agrosystems Geosci. Environ.* 3, e20105. <https://doi.org/10.1002/agg2.20105>.
- Haruna, S.I., Eichas, R.C., Peters, O.M., Farmer, A.C., Lackey, D.Q., Nichols, J.E., Peterson, W.H., Slone, N.A., 2022. In situ water infiltration: influence of cover crops after growth termination. *Soil Sci. Soc. Am. J.* 86, 769–780. <https://doi.org/10.1002/saj2.20390>.
- Haruna, S.I., Eke, D., Haque, MdA., Ku, S., 2025. Cover crop effects on select soil physicochemical and biological properties. *Agric. Ecosyst. Environ.* 393, 109817. <https://doi.org/10.1016/j.agee.2025.109817>.
- Haynes, R.J., Beare, M.H., 1997. Influence of six crop species on aggregate stability and some labile organic matter fractions. *Soil Biol. Biochem.* 29, 1647–1653. [https://doi.org/10.1016/S0038-0717\(97\)00078-3](https://doi.org/10.1016/S0038-0717(97)00078-3).
- Henry, A., Kleinman, P.J.A., Lynch, J.P., 2009. Phosphorus runoff from a phosphorus deficient soil under common bean (*Phaseolus vulgaris* L.) and soybean (*Glycine max* L.) genotypes with contrasting root architecture. *Plant Soil* 317, 1–16. <https://doi.org/10.1007/s11104-008-9784-0>.
- Hernández, A.J., Lacasta, C., Pastor, J., 2005. Effects of different management practices on soil conservation and soil water in a rainfed olive orchard. *Agric. Water Manag., Special Issue on Land and Water Use: Environmental Management Tools and Practices* 77, 232–248. <https://doi.org/10.1016/j.agwat.2004.09.030>.
- Hinsinger, P., Bengough, A.G., Vetterlein, D., Young, I.M., 2009. Rhizosphere: biophysics, biogeochemistry and ecological relevance. *Plant Soil* 321, 117–152. <https://doi.org/10.1007/s11104-008-9885-9>.
- Hu, Y., Zhang, F., Luo, Z., Badreldin, N., Benoy, G., Xing, Z., 2023. Soil and water conservation effects of different types of vegetation cover on runoff and erosion driven by climate and underlying surface conditions. *Catena* 231, 107347. <https://doi.org/10.1016/j.catena.2023.107347>.
- Huang, J., Zhao, X., Wu, P., 2013. Surface runoff volumes from vegetated slopes during simulated rainfall events. *J. Soil Water Conserv.* 68, 283–295. <https://doi.org/10.2489/jswc.68.4.283>.
- Huang, Z., Tian, F., Wu, G., Liu, Y., Dang, Z., 2017. Legume grasslands promote precipitation infiltration better than graminaceous grasslands in arid regions. *Land Degrad. Dev.* 28, 309–316. <https://doi.org/10.1002/ldr.2635>.
- Iderawumi, A.M., Kamal, T.O., 2022. Green manure for agricultural sustainability and improvement of soil fertility. *Farming Manag.* 7. <https://doi.org/10.31830/2456-8724.2022.FM-101>.
- Iglesias, I., Lloveras, J., 1998. Annual cool-season legumes for forage production in mild winter areas. *Grass Forage Sci.* 53, 318–325. <https://doi.org/10.1046/j.1365-2494.1998.00135.x>.
- Jia, Y., Huan, H., Zhang, W., Wan, B., Sun, J., Tu, Z., 2024. Soil infiltration mechanisms under plant root disturbance in arid and semi-arid grasslands and the response of solute transport in rhizosphere soil. *Sci. Total Environ.* 957, 177633. <https://doi.org/10.1016/j.scitotenv.2024.177633>.
- Jian, J., Du, X., Stewart, R.D., 2020a. Quantifying cover crop effects on soil health and productivity. *Data Brief* 29, 105376. <https://doi.org/10.1016/j.dib.2020.105376>.
- Jian, J., Lester, B.J., Du, X., Reiter, M.S., Stewart, R.D., 2020b. A calculator to quantify cover crop effects on soil health and productivity. *Soil Tillage Res.* 199, 104575. <https://doi.org/10.1016/j.still.2020.104575>.
- Jiang, J., Weiss, M., Liu, S., Baret, F., 2022. Effective GAI is best estimated from reflectance observations as compared to GAI and LAI: demonstration for wheat and maize crops based on 3D radiative transfer simulations. *Field Crops Res.* 283, 108538. <https://doi.org/10.1016/j.fcr.2022.108538>.
- Jones, D.L., Nguyen, C., Finlay, R.D., 2009. Carbon flow in the rhizosphere: carbon trading at the soil–root interface. *Plant Soil* 321, 5–33. <https://doi.org/10.1007/s11104-009-9925-0>.
- Kabalan, S., Juhos, K., Tóth, E., Biró, B., 2022. Impact of symbiotic mycorrhiza interrelation in some soil biological parameters and growth of five cover crops. *Agrokém. És Talajt.* 71, 135–147. <https://doi.org/10.1556/0088.2022.00117>.
- Kaye, J.P., Quemada, M., 2017. Using cover crops to mitigate and adapt to climate change. A review. *Agron. Sustain. Dev.* 37, 4. <https://doi.org/10.1007/s13593-016-0410-x>.
- Kervroëdan, L., Armand, R., Rey, F., Faucon, M.-P., 2021. Trait-based sediment retention and runoff control by herbaceous vegetation in agricultural catchments: a review. *Land Degrad. Dev.* 32, 1077–1089. <https://doi.org/10.1002/ldr.3812>.
- Khan, F.U., Khan, A.A., Li, K., Xu, X., Adnan, M., Fahad, S., Ahmad, R., Khan, M.A., Nawaz, T., Zaman, F., 2022. Influences of long-term crop cultivation and fertilizer management on soil aggregates stability and fertility in the Loess Plateau, Northern China. *J. Soil Sci. Plant Nutr.* 22, 1446–1457. <https://doi.org/10.1007/s42729-021-00744-1>.
- Kinnell, P.I.A., 2022. Insights into the sediment transport processes operating in rain-impacted flows. *Catena* 217, 106452. <https://doi.org/10.1016/j.catena.2022.106452>.
- Kocira, A., Staniak, M., Tomaszewska, M., Kornas, R., Cymerman, J., Panasiewicz, K., Lipińska, H., 2020. Legume cover crops as one of the elements of strategic weed management and soil quality improvement. A review. *Agriculture* 10, 394. <https://doi.org/10.3390/agriculture10090394>.
- Koudahe, K., Allen, S.C., Djaman, K., 2022. Critical review of the impact of cover crops on soil properties. *Int. Soil Water Conserv. Res.* 10, 343–354. <https://doi.org/10.1016/j.iswcr.2022.03.003>.
- Kumawat, A., Bamboriya, S.D., Meena, R.S., Yadav, D., Kumar, A., Kumar, S., Raj, A., Pradhan, G., 2022. Legume-based inter-cropping to achieve the crop, soil, and environmental health security. In: *Advances in Legumes for Sustainable Intensification*. Elsevier, pp. 307–328. <https://doi.org/10.1016/B978-0-323-85797-0.00005-7>.
- Kumawat, A., Yadav, D., Srivastava, P., Babu, S., Kumar, D., Singh, D., Vishwakarma, D. K., Sharma, V.K., Madhu, M., 2023. Restoration of agroecosystems with conservation agriculture for food security to achieve sustainable development goals. *Land Degrad. Dev.* 34, 3079–3097. <https://doi.org/10.1002/ldr.4677>.
- Li, L., Chen, P., Wang, K., Zhang, R., Yuan, X., Ge, L., Li, Q., Liu, Y., Zhang, X., Li, Z., 2023. Gramineae-legumes mixed planting effectively reduces soil and nutrient loss in orchards. *Agric. Water Manag.* 289, 108513. <https://doi.org/10.1016/j.agwat.2023.108513>.
- Liu, S., Dong, A., Niu, B., Xu, F., Xu, J., Yin, L., Wang, S., 2025. Maize/cover crop intercropping mitigates soil erosion and enhances yield of ridge cultivation in Chinese Mollisol region. *Catena* 255, 109012. <https://doi.org/10.1016/j.catena.2025.109012>.
- Liu, Y., Guo, L., Huang, Z., López-Vicente, M., Wu, G.-L., 2020. Root morphological characteristics and soil water infiltration capacity in semi-arid artificial grassland soils. *Agric. Water Manag.* 235, 106153. <https://doi.org/10.1016/j.agwat.2020.106153>.
- Liu, Y.-F., Liu, Y., Wu, G.-L., Shi, Z.-H., 2019. Runoff maintenance and sediment reduction of different grasslands based on simulated rainfall experiments. *J. Hydrol.* 572, 329–335. <https://doi.org/10.1016/j.jhydrol.2019.03.008>.
- Ma, D., Yin, L., Ju, W., Li, X., Liu, X., Deng, X., Wang, S., 2021. Meta-analysis of green manure effects on soil properties and crop yield in northern China. *Field Crops Res.* 266, 108146. <https://doi.org/10.1016/j.fcr.2021.108146>.
- Mantino, A., Ragagnini, G., Nassi O Di Nasso, N., Tozzini, C., Taccini, F., Bonari, E., 2016. Alfalfa (*Medicago sativa* L.) overseeding on mature switchgrass (*Panicum virgatum* L.) stand: biomass yield and nutritive value after the establishment year. *Ital. J. Agron.* 11, 747. <https://doi.org/10.4081/ija.2016.747>.
- Marley, C.L., Fychan, R., Davies, J.W., Scott, M., Crotty, F.V., Sanderson, R., Scullion, J., 2024. Grasslands and flood mitigation – contrasting forages improve surface water infiltration rates. *Sci. Total Environ.* 951, 175598. <https://doi.org/10.1016/j.scitotenv.2024.175598>.
- McCallum, M.H., Kirkegaard, J.A., Green, T.W., Cresswell, H.P., Davies, S.L., Angus, J.F., Peoples, M.B., 2004. Improved subsoil macroporosity following perennial pastures. *Aust. J. Exp. Agric.* 44, 299–307. <https://doi.org/10.1071/EA03076>.
- Meng, L., Ma, Y., Cheng, T., Gignac, J., Guan, C., Wu, G.-L., 2025. Gramineae-Legumes-mixed cultivated grassland improves soil anti-erodibility of the steep loess slopes in semi-arid region. *J. Environ. Manag.* 391, 126489. <https://doi.org/10.1016/j.jenvman.2025.126489>.
- Meyer, L.D., Wischmeier, W.H., 1969. Mathematical simulation of the process of soil erosion by water. *Trans ASAE (Am. Soc. Agric. Eng.)* 12, 754–758. <https://doi.org/10.13031/2013.38945>.
- Mohammed, S., Hassan, E., Abdo, H.G., Szabo, S., Mokhtar, A., Alsafadi, K., Al-Khouri, I., Rodrigo-Comino, J., 2021. Impacts of rainstorms on soil erosion and organic matter for different cover crop systems in the Western Coast agricultural region of Syria. *Soil Use Manag.* 37, 196–213. <https://doi.org/10.1111/sum.12683>.
- Molla, A., Skoufogianni, E., Lolas, A., Skordas, K., 2022. The impact of different cultivation practices on surface runoff, soil and nutrient losses in a rotational system of legume–cereal and sunflower. *Plants* 11, 3513. <https://doi.org/10.3390/plants11243513>.
- Morgan, R.P.C., Quinton, J.N., Smith, R.E., Govers, G., Poesen, J.W.A., Auerswald, K., Chisci, G., Torri, D., Styczen, M.E., Folly, A.J.V., 1998. The European Soil Erosion

- Model (EUROSEM): Documentation and User Guide. Silsoe College, Cranfield University, Silsoe, UK.
- Muñoz-Romero, V., López-Bellido, L., López-Bellido, R.J., 2011. Faba bean root growth in a vertisol: tillage effects. *Field Crops Res.* 120, 338–344. <https://doi.org/10.1016/j.fcr.2010.11.008>.
- Murphy-Bokern, D., Stoddard, F., Watson, C. (Eds.), 2017. *Legumes in Cropping Systems*, CABI Books. CABI, Oxfordshire Boston, MA. <https://doi.org/10.1079/9781780644981.0000>.
- Mwanamwenge, J., Siddique, K.H.M., Sedgley, R.H., 1997. Canopy development and light absorption of grain legume species in a short season mediterranean-type environment. *J. Agron. Crop Sci.* 179, 1–7. <https://doi.org/10.1111/j.1439-037X.1997.tb01141.x>.
- Naderi Marangal, N., Sadeghi, S.H., Erfanzadeh, R., Khaledi Darvishan, A., 2025. Deciphering runoff and soil loss processes during various phenological stages of *Trifolium Pratense* at small plot scale. *Soil Sci. Soc. Am. J.* 89, e70100. <https://doi.org/10.1002/saj2.70100>.
- Neuschwandtner, R.W., Bernhuber, A., Kammlander, S., Wagentristsl, H., Klimek-Kopyra, A., Kaul, H.-P., 2019. Agronomic potential of winter grain legumes for Central Europe: development, soil coverage and yields. *Field Crops Res.* 241, 107576. <https://doi.org/10.1016/j.fcr.2019.107576>.
- Novara, A., Gristina, L., Saladino, S.S., Santoro, A., Cerdà, A., 2011. Soil erosion assessment on tillage and alternative soil managements in a Sicilian vineyard. *Soil Tillage Res.* 117, 140–147. <https://doi.org/10.1016/j.still.2011.09.007>.
- Pala, M., Armstrong, E., Johansen, C., 2000. The role of legumes in sustainable cereal production in rainfed areas. In: Knight, R. (Ed.), *Linking Research and Marketing Opportunities for Pulses in the 21st Century*, Current Plant Science and Biotechnology in Agriculture. Springer, Netherlands, Dordrecht, pp. 323–334. https://doi.org/10.1007/978-94-011-4385-1_29.
- Panagos, P., Ballabio, C., Borrelli, P., Meusburger, K., Klik, A., Rousseva, S., Tadić, M.P., Michaelides, S., Hrabalíková, M., Olsen, P., Aalto, J., Lakatos, M., Rymaszewicz, A., Dumitrescu, A., Begueria, S., Alewell, C., 2015a. Rainfall erosivity in Europe. *Sci. Total Environ.* 511, 801–814. <https://doi.org/10.1016/j.scitotenv.2015.01.008>.
- Panagos, P., Borrelli, P., Meusburger, K., Alewell, C., Lugato, E., Montanarella, L., 2015b. Estimating the soil erosion cover-management factor at the European scale. *Land Use Policy* 48, 38–50. <https://doi.org/10.1016/j.landusepol.2015.05.021>.
- Panagos, P., Matthews, F., Patault, E., De Michele, C., Quaranta, E., Bezak, N., Kaffas, K., Patro, E.R., Auel, C., Schleiss, A.J., Fendrich, A., Liakos, L., Van Eynde, E., Vieira, D., Borrelli, P., 2024. Understanding the cost of soil erosion: an assessment of the sediment removal costs from the reservoirs of the European Union. *J. Clean. Prod.* 434, 140183. <https://doi.org/10.1016/j.jclepro.2023.140183>.
- Pimentel, D., Burgess, M., 2013. Soil erosion threatens food production. *Agriculture* 3, 443–463. <https://doi.org/10.3390/agriculture3030443>.
- Qu, Y., Wu, Q., Khan, F.U., Wang, J., Ren, X., Chai, X., Xu, X., Du, F., 2025. The impact of root systems on soil macropore abundance and soil infiltration capacity. *Plant Soil*. <https://doi.org/10.1007/s11104-025-07237-5>.
- Quintarelli, V., Radicetti, E., Allevato, E., Stazi, S.R., Haider, G., Abideen, Z., Bibi, S., Jamal, A., Mancinelli, R., 2022. Cover crops for sustainable cropping systems: a review. *Agriculture* 12, 2076. <https://doi.org/10.3390/agriculture12122076>.
- Quinton, J.N., Fiener, P., 2024. Soil erosion on arable land: an unresolved global environmental threat. *Prog. Phys. Geogr. Earth Environ.* 48, 136–161. <https://doi.org/10.1177/03091333231216595>.
- Ramirez-Garcia, J., Gabriel, J.L., Alonso-Ayuso, M., Quemada, M., 2015. Quantitative characterization of five cover crop species. *J. Agric. Sci.* 153, 1174–1185. <https://doi.org/10.1017/S0021859614000811>.
- Ramsankaran, R., Kothiyari, U.C., Ghosh, S.K., Malcherek, A., Murugesan, K., 2013. Physically-based distributed soil erosion and sediment yield model (DREAM) for simulating individual storm events. *Hydrol. Sci. J.* 58, 872–891. <https://doi.org/10.1080/02626667.2013.781606>.
- Reichmann, M., Blanc, L., Lampurlanés, J., Simon-Miquel, G., Plaza-Bonilla, D., 2025. Does intercropping improve soil aggregation and organic carbon protection? A case-study in the semi-arid Mediterranean. *Agric. Ecosyst. Environ.* 385, 109563. <https://doi.org/10.1016/j.agee.2025.109563>.
- Renard, K.G., Foster, G.R., Weesies, G.A., McCool, D.K., Yoder, D.C., 1997. *Predicting Soil Erosion by Water: a Guide to Conservation Planning with the Revised Universal Soil Loss Equation (RUSLE)* (Agriculture Handbook No. 703). U.S. Department of Agriculture, Washington, DC.
- Rodríguez, B.C., Zuazo, V.H.D., Rodríguez, M.S., García-Tejero, I.F., Ruiz, B.G., De Torres, M.A.R.-R., Ordóñez-Fernández, R., Carbonell-Bojollo, R.M., Távira, S.C., 2022. Legumes protect the soil erosion and ecosystem services. In: *Advances in Legumes for Sustainable Intensification*. Elsevier, pp. 247–266. <https://doi.org/10.1016/B978-0-323-85797-0.00010-0>.
- Rodrigo-Comino, J., Terol, E., Mora, G., Giménez-Morera, A., Cerdà, A., 2020. Vicia sativa Roth. Can reduce soil and water losses in recently planted vineyards (Vitis vinifera L.). *Earth Syst. Environ.* 4, 827–842. <https://doi.org/10.1007/s41748-020-00191-5>.
- Sastre, B., Barbero-Sierra, C., Bienes, R., Marques, M.J., García-Díaz, A., 2017. Soil loss in an olive grove in Central Spain under cover crops and tillage treatments, and farmer perceptions. *J. Soils Sediments* 17, 873–888. <https://doi.org/10.1007/s11368-016-1589-9>.
- Sastre, B., Marques, M.J., García-Díaz, A., Bienes, R., 2018. Three years of management with cover crops protecting sloping olive groves soils, carbon and water effects on gypsiferous soil. *Catena* 171, 115–124. <https://doi.org/10.1016/j.catena.2018.07.003>.
- Scavo, A., Fontanazza, S., Restuccia, A., Pesce, G.R., Abbate, C., Mauromicale, G., 2022. The role of cover crops in improving soil fertility and plant nutritional status in temperate climates. A review. *Agron. Sustain. Dev.* 42, 93. <https://doi.org/10.1007/s13593-022-00825-0>.
- Sharma, P., Singh, A., Kahlon, C.S., Brar, A.S., Grover, K.K., Dia, M., Steiner, R.L., 2018. The role of cover crops towards sustainable soil health and agriculture—A review paper. *Am. J. Plant Sci.* 9, 1935–1951. <https://doi.org/10.4236/ajps.2018.99140>.
- Silvestroni, O., Dottori, E., Pallotti, L., Lattanzi, T., Santilocchi, R., Lanari, V., 2024. Using legume-enriched cover crops to improve grape yield and quality in hillside Vineyards. *Agronomy* 14, 2528. <https://doi.org/10.3390/agronomy14112528>.
- Singh Jatav, H., Rajput, V.D. (Eds.), 2022. *Ecosystem Services: Types, Management and Benefits, Agriculture Issues and Policies*. Nova Science Publishers, New York.
- Skaalsveen, K., Ingram, J., Clarke, L.E., 2019. The effect of no-till farming on the soil functions of water purification and retention in north-western Europe: a literature review. *Soil Tillage Res.* 189, 98–109. <https://doi.org/10.1016/j.still.2019.01.004>.
- Sun, W., Shahrajabian, M.H., 2022. The effectiveness of Rhizobium bacteria on soil fertility and sustainable crop production under cover and catch crops management and green manuring. *Not. Bot. Horti Agrobot. Cluj-Napoca* 50, 12560. <https://doi.org/10.15835/nbha50212560>.
- Thomas, D.T., Moore, A.D., Bell, L.W., Webb, N.P., 2018. Ground cover, erosion risk and production implications of targeted management practices in Australian mixed farming systems: lessons from the grain and graze program. *Agric. Syst.* 162, 123–135. <https://doi.org/10.1016/j.agry.2018.02.001>.
- Thomas, G.A., Orange, D.N., King, A.J., 2008. Effects of crop and pasture rotations and surface cover on rainfall infiltration on a Kandosol in south-west Queensland. *Soil Res.* 46, 203–209. <https://doi.org/10.1017/SR07162>.
- Todisco, F., Massimi Alunno, A., Vergni, L., 2025. Spatial distribution, temporal behaviour, and trends of rainfall erosivity in central Italy using coarse data. *Water* 17, 801. <https://doi.org/10.3390/w17060801>.
- Todisco, F., Vergni, L., Ceppitelli, R., 2023. Modelling the dynamics of seal formation and pore clogging in the soil and its effect on infiltration using membrane fouling models. *J. Hydrol.* 618, 129208. <https://doi.org/10.1016/j.jhydrol.2023.129208>.
- Tomić, D.D., Stevović, V.I., Đurović, D., Marjanović, M., Madić, M.R., Pavlović, N., Lazarević, D., Petrović, M., Radovanović, M., 2023. Perennial forage legumes as an element of sustainable systems. *Not. Bot. Horti Agrobot. Cluj-Napoca* 51, 13240. <https://doi.org/10.15835/nbha51313240>.
- Vallebona, C., Mantino, A., Bonari, E., 2016. Exploring the potential of perennial crops in reducing soil erosion: a GIS-based scenario analysis in southern Tuscany, Italy. *Appl. Geogr.* 66, 119–131. <https://doi.org/10.1016/j.apgeog.2015.11.015>.
- Van Eerd, L.L., Chahal, I., Peng, Y., Awrey, J.C., 2023. Influence of cover crops at the four spheres: a review of ecosystem services, potential barriers, and future directions for North America. *Sci. Total Environ.* 858, 159990. <https://doi.org/10.1016/j.scitotenv.2022.159990>.
- Wagg, C., van Erck, A., Fava, E., Comeau, L.-P., Mitterboeck, T.F., Goyer, C., Li, S., McKenzie-Gopsill, A., Mills, A., 2021. Full-season cover crops and their traits that promote agroecosystem services. *Agriculture* 11, 830. <https://doi.org/10.3390/agriculture11090830>.
- Wang, J., Liu, G., Yang, Y., Wang, B., 2026. Construction of a comprehensive root parameter to reflect the effect of plant root systems on soil erosion. *Land Degrad. Dev.* <https://doi.org/10.1002/ldr.70456>.
- Wischmeier, W.H., Smith, D.D., 1978. *Predicting Rainfall Erosion Losses: a Guide to Conservation Planning* (Agriculture Handbook No. 537). Department of Agriculture, Science and Education Administration, Washington, DC.
- Wong, K.M., Griffiths, M., Moran, A., Johnston, A., Liu, A.E., Sellers, M.A., Topp, C.N., 2024. Cover crop cultivars and species differ in root traits potentially impacting their selection for ecosystem services. *Plant Soil* 500, 279–296. <https://doi.org/10.1007/s11104-023-06431-7>.
- Wu, G.-L., Huang, Z., Liu, Y.-F., Cui, Z., Liu, Y., Chang, X., Tian, F.-P., López-Vicente, M., Shi, Z.-H., 2019. Soil water response of plant functional groups along an artificial legume grassland succession under semi-arid conditions. *Agric. For. Meteorol.* 278, 107670. <https://doi.org/10.1016/j.agrformet.2019.107670>.
- Wu, G.-L., Yang, Z., Cui, Z., Liu, Y., Fang, N.-F., Shi, Z.-H., 2016. Mixed artificial grasslands with more roots improved mine soil infiltration capacity. *J. Hydrol.* 535, 54–60. <https://doi.org/10.1016/j.jhydrol.2016.01.059>.
- Wu, S., Wu, P., Feng, H., Merkley, G.P., 2011. Effects of alfalfa coverage on runoff, erosion and hydraulic characteristics of overland flow on loess slope plots. *Front. Environ. Sci. Eng. China* 5, 76–83. <https://doi.org/10.1007/s11783-011-0282-x>.
- Xu, B., Shan, L., Li, F., Jiang, J., 2007. Seasonal and spatial root biomass and water use efficiency of four forage legumes in semiarid northwest China. *Afr. J. Biotechnol.* 6, 2708–2714. <https://doi.org/10.5897/AJB2007.000-2433>.
- Yakupoglu, T., Basaran, U., Ozdemir, N., Acar, Z., 2011. Quantifying aggregate stability of a clay soil under annual forage crops. *J. Plant Nutr. Soil Sci.* 174, 404–407. <https://doi.org/10.1002/jpln.200800332>.
- Yao, C., Ye, S., Chen, S., Zhang, M., Zhu, M., Wu, F., 2025. The effect of soybean growth on the seasonal variation of hillslopes erosion in gently sloping farmland on the Loess Plateau. *Land Degrad. Dev.* 36, 3975–3988. <https://doi.org/10.1002/ldr.5611>.
- Yüksek, F., Yüksek, T., 2015. Growth performance of Sainfoin and its effects on the runoff, soil loss and sediment concentration in a semi-arid region of Turkey. *Catena* 133, 309–317. <https://doi.org/10.1016/j.catena.2015.05.018>.

- Yu, Y., Loiskandl, W., Kaul, H.-P., Himmelbauer, M., Wei, W., Chen, L., Bodner, G., 2016. Estimation of runoff mitigation by morphologically different cover crop root systems. *J. Hydrol.* 538, 667–676. <https://doi.org/10.1016/j.jhydrol.2016.04.060>.
- Zhang, Y., Dai, X., Yang, K., Liu, C., Wang, M., Yan, F., Chen, X., Wang, E., 2026. From physical regulation to biochemical dominance: erosion-driven shifts in plant regulation of aggregate stability in Mollisols. *Catena* 266, 109923. <https://doi.org/10.1016/j.catena.2026.109923>.
- Zhao, X., Mak-Mensah, E., Zhao, W., Wang, Q., Zhou, X., Zhang, D., Zhu, J., Qi, W., Liu, Q., Xiaoling, L., Xuchun, L., Liu, B., 2024. Optimized ridge-furrow technology with biochar amendment for alfalfa yield enhancement and soil erosion reduction based on a structural equation model on sloping land. *Agric. Water Manag.* 298, 108866. <https://doi.org/10.1016/j.agwat.2024.108866>.
- Zuazo, V.H.D., Rodríguez, B.C., García-Tejero, I.F., Ruiz, B.G., Távira, S.C., 2020. Benefits of organic olive rainfed systems to control soil erosion and runoff and improve soil health restoration. *Agron. Sustain. Dev.* 40, 41. <https://doi.org/10.1007/s13593-020-00644-1>.